

PROBLEM SOLVING
**THE POSSIBILITIES, LIMITATIONS,
AND HISTORY OF USING SCIENCE
TO SOLVE OUR PROBLEMS**

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ISSUE #4

PROVING KEPLER'S LAW

THE MATHEMATICS OF
IMPOSSIBILITY

PIEZOELECTRIC ENERGY
HARVESTING FOR NEXT-GENERATION
AIRCRAFT AND AVIONIC SYSTEMS

The Science



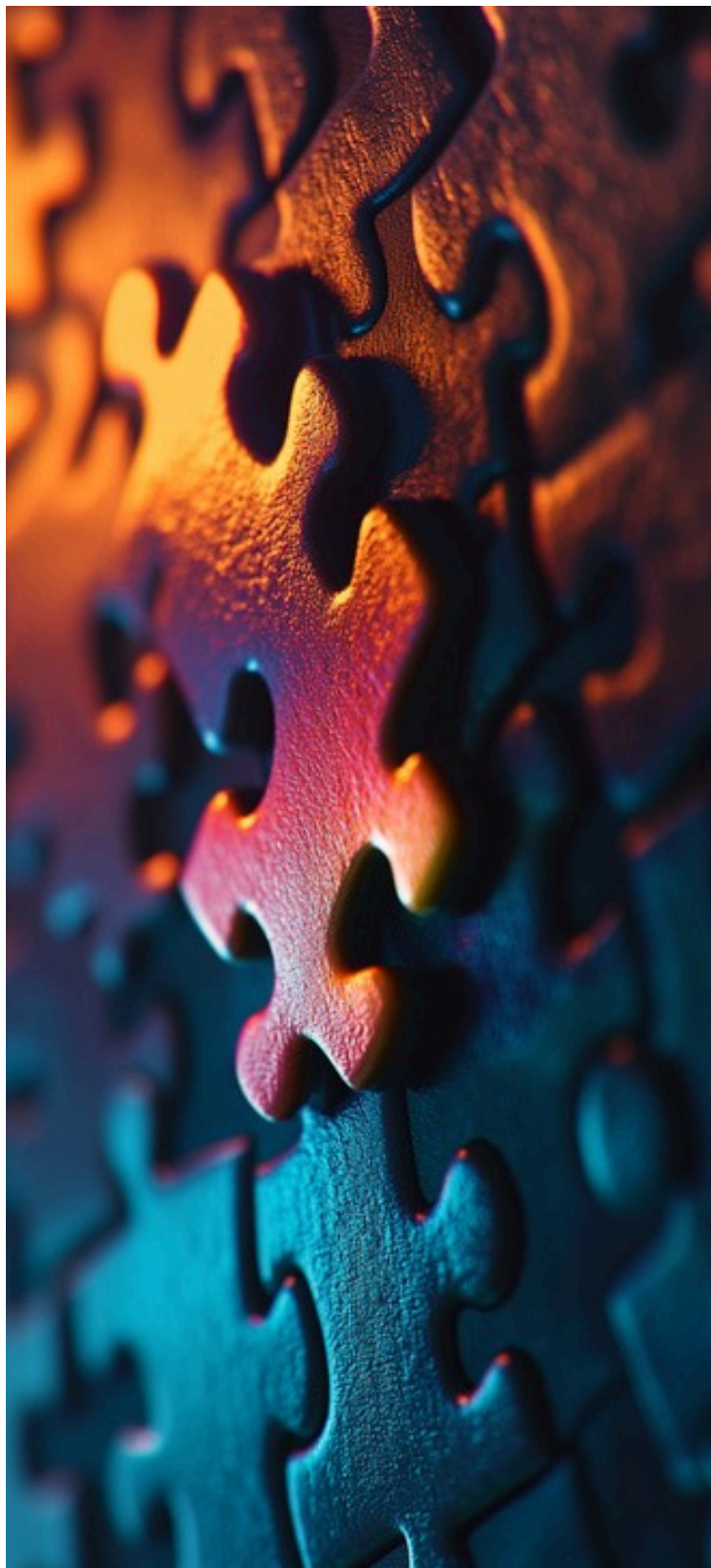
of Solving

THE CENTRIFUGE

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Problem solving is ingrained in every aspect of our lives. Every day, we are confronted with decisions, trade-offs and suboptimal situations which vary in magnitude and complexity. From habit adjustments to life-changing decisions, solving problems and knowing how to come up with effective solutions is an essential skill to develop, and one that is beneficial in our personal and professional lives.

What we sought to uncover in this issue revolved around this concept. We asked ourselves three questions: how have we been problem-solving in STEM; how effective are these approaches; and how do they connect across disciplines?

The primary objective of most mathematics, engineering and science courses is to solve problems. It is to teach students how to perfect existing solutions and to question the status quo.

In this issue, we bring you articles on some of the most innovative applications of this face of science, from how mathematics has set its limitations and when proof shows us that math may not be the complete problem-solving skill. You will learn about the most recent innovation in aviation which aim to maximize energy recovery in avionic systems. You will be able to explore the same drive for improvement in the search on how pesticides can affect core physiological activities and much more.

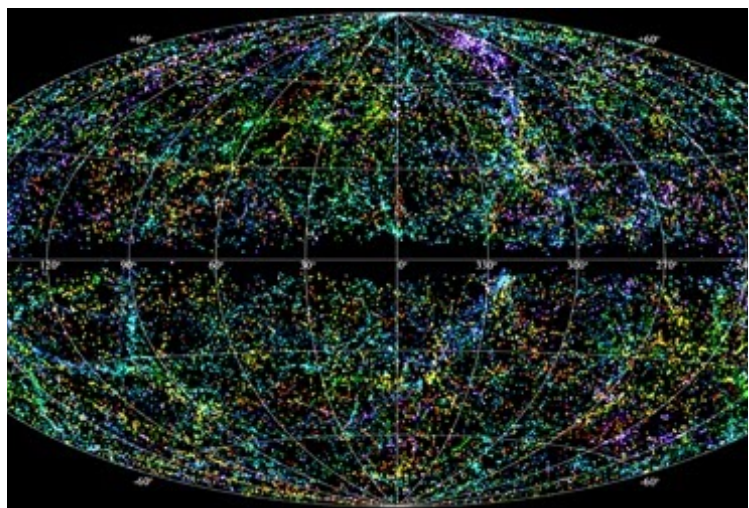
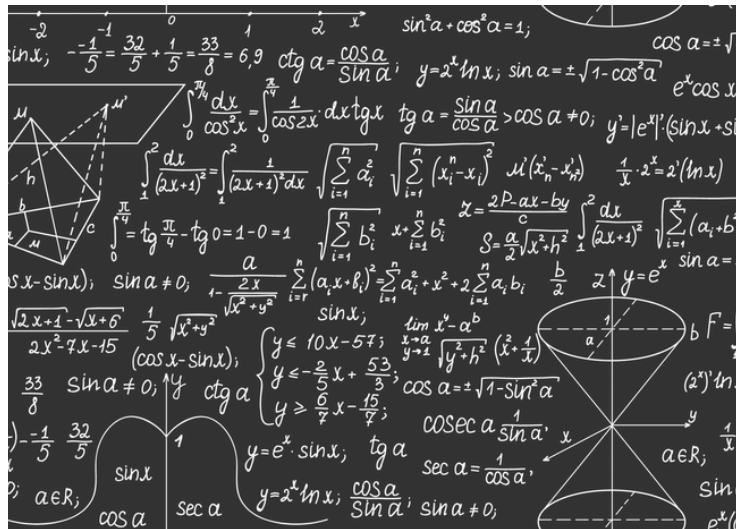
By learning how we can use our knowledge and scientific skills to improve real-world systems, we can be better prepared to face barriers, not only with creativity, but with a deeply empirical and rational mindset.

OUR WRITERS



RITISHA AGARWAL

HAILING FROM LONDON, RITISHA IS DRIVEN BY THE BELIEF THAT MATHEMATICS IS NOT JUST AN ABSTRACT LANGUAGE, BUT A PRACTICAL LENS THROUGH WHICH WE CAN DECODE THE WORLD. SHE IS PASSIONATE ABOUT THE REAL-WORLD APPLICATION OF MATHEMATICAL CONCEPTS, CONSTANTLY SEEKING TO BRIDGE THE GAP BETWEEN THEORY AND EVERYDAY LIFE. A LIFELONG LEARNER AT HEART, SHE THRIVES ON EXPLORING DIVERSE TOPICS AND CONNECTING WITH A GLOBAL COMMUNITY OF THINKERS WHO SHARE HER RELENTLESS DRIVE FOR DISCOVERY.



IBRAHIM CHISHTI

IBRAHIM CHISHTI IS A SIXTH FORM STUDENT FROM LONDON. HE IS FASCINATED BY THE UNIVERSE WE LIVE IN AND THE WAYS THAT WE QUANTIFY AND UNDERSTAND IT. WITH A PARTICULAR PASSION FOR ASTRONOMY AND ENGINEERING, HE IS DRIVEN BY THE IDEA OF HUMAN EXPLORATION BEYOND EARTH AND HOPES TO ONE DAY HELP WITH SUCH EXPLORATION. HE ENJOYS BOTH LEARNING ABOUT AND SHARING KNOWLEDGE ON OUR PLACE IN THIS VERY BIG WORLD.

PUNEETH NUNNA

PUNEETH NUNNA IS A STUDENT AT SHREWSBURY SENIOR HIGH SCHOOL WHO IS PASSIONATE ABOUT THE FOREFRONT OF MODERN TECHNOLOGY, PARTICULARLY THE EVOLVING INTERSECTION OF THE PHYSICAL AND DIGITAL WORLDS. AS TOOLS SUCH AS ARTIFICIAL INTELLIGENCE AND ADVANCED HARDWARE BECOME MORE ACCESSIBLE, HE STRIVES TO USE THEM FOR MEANINGFUL, ORIGINAL RESEARCH AND TO CREATE RESOURCES THAT LET OTHERS BUILD ON HIS WORK. HIS INTERESTS SPAN ROBOTICS AND ENGINEERING TO ARTIFICIAL INTELLIGENCE AND ENVIRONMENTAL SUSTAINABILITY, AND HE IS EXCITED TO KEEP EXPLORING THESE FIELDS THROUGH RESEARCH, INNOVATION, AND COLLABORATION.





UNNATI MISHRA

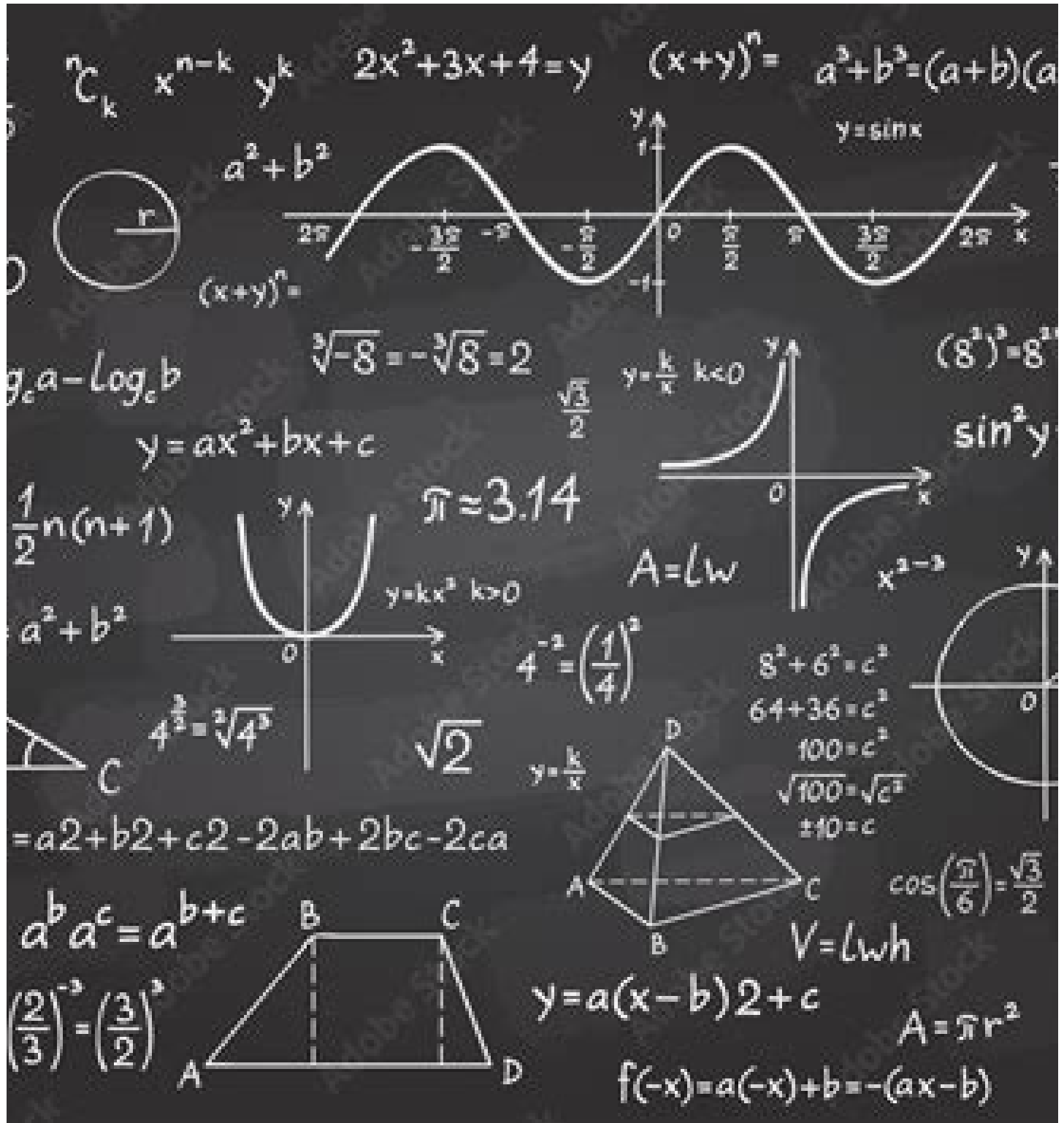
UNNATI MISHRA IS A RISING BIOMEDICAL ENGINEERING STUDENT AT THE UNIVERSITY OF SOUTHERN CALIFORNIA. THEY DEVELOPED A PASSION FOR RESEARCH IN HIGH SCHOOL THROUGH INDEPENDENT AND GROUP PROJECTS, PARTICULARLY IN EXPLORING THE EFFECTS OF ENVIRONMENTAL CHEMICALS ON HUMAN HEALTH. THEY ARE ESPECIALLY INTERESTED IN THE INTERSECTION OF BIOMEDICAL ENGINEERING AND GLOBAL HEALTH POLICY, AND HOPE TO CONTINUE CONDUCTING RESEARCH THAT IMPROVES HEALTHCARE ACCESSIBILITY. WHEN NOT IN THE LAB,



THE MATHEMATICS OF IMPOSSIBILITY: WHEN PROOF SHOWS US THAT SOME THINGS CANNOT BE DONE



By Ritisha Agarwal





ABSTRACT

For thousands of years, mathematics was viewed as the ultimate problem-solving machine. A question was asked, a clever person searched long enough, and eventually an answer emerged. Then mathematics discovered something far stranger: sometimes the answer does not exist.

This article explores the mathematics of impossibility: from ancient geometric constructions that cannot be completed, to modern discoveries showing that some mathematical truths cannot be proven, some computational problems cannot be solved, and some futures cannot be predicted.

The story of impossibility is not a story of failure.

It is the story of mathematics becoming powerful enough to understand its own limits.

Introduction: How Mathematics Discovered the Boundaries of Human Knowledge

For thousands of years, mathematics was viewed as the ultimate problem-solving machine. A question was asked, a clever person searched long enough, and eventually an answer emerged. The universe seemed like a locked box, but mathematics was the key.

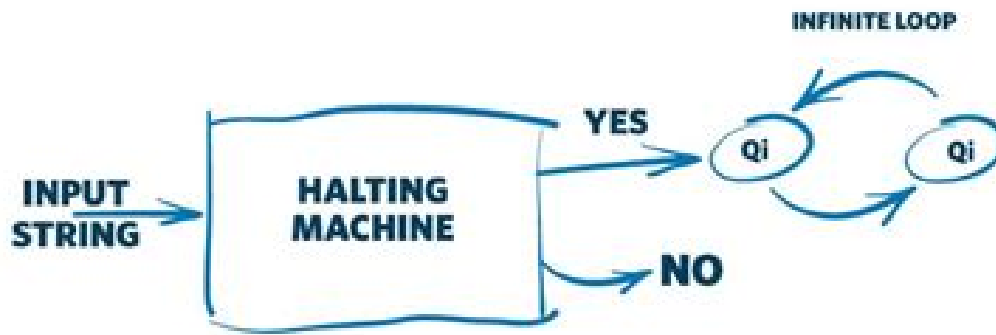
Then mathematics discovered something far stranger: sometimes the key does not exist. Not because humans are not clever enough. Not because we have not searched long enough. But because the thing we are searching for cannot exist at all.

Mathematics, the subject built around finding answers, became the subject that could prove there were no answers. This is one of the most beautiful contradictions in mathematics: the ability to prove failure. A mathematician does not merely say "I tried many methods and nothing worked." That is not a mathematical argument. Instead, mathematics can say: "No method will ever work, because the rules of the universe make the solution impossible."

This idea appears everywhere:

- Ancient problems of doubling the cube, trisecting the angle, and squaring the circle.
- Algebra showing that certain equations have no solutions.
- Kurt Gödel proving that some truths cannot be proven within a formal system.
- Alan Turing proving that some problems cannot be solved by any computer.

The story of impossibility is not a story of defeat. It is a story of mathematics becoming powerful enough to understand its own boundaries.



Graphic representation of the halting problem - an impossible problem for algorithms to resolve

THE FIRST GREAT IMPOSSIBLE PROBLEMS THE GREEKS AND THE GEOMETRY OF PERFECTION

More than two thousand years ago, ancient Greek mathematicians believed geometry was the language of the universe. Shapes were not just drawings. They represented mathematical ideals. A circle was not simply a round object. It was the perfect collection of all points exactly the same distance from a centre. A straight line was not just something drawn with a ruler. It represented infinite precision.

The Greeks created a beautiful system: Given certain tools, especially a compass and an unmarked straightedge, one could construct certain geometric objects. A compass allows you to draw perfect circles. A straightedge allows you to draw perfect straight lines. Together, these two tools created an entire universe of possible constructions. But mathematicians eventually discovered something astonishing: some shapes were outside this universe. The tools themselves created limits.

The Three Famous Impossible Constructions

Three problems became legendary:

1. Squaring the circle.
2. Doubling the cube.
3. Trisecting an angle.

For centuries, mathematicians tried to solve them. For centuries, they failed. Eventually, they discovered something astonishing: the problems were not difficult. They were impossible.



Squaring the Circle: Can a Circle Become a Square?

Imagine drawing a perfect circle. Now imagine being asked: Construct a square with exactly the same area as this circle. At first this sounds reasonable.

A circle has an area:

$$A = \pi r^2$$

A square has an area:

$$A = S^2$$

So if the circle has radius , the square must satisfy:

$$S^2 = \pi r^2$$

Taking square roots:

$$S = r\sqrt{\pi}$$

So the problem becomes: Can we construct a length equal to $\sqrt{\pi}$ using only a compass and straightedge? For thousands of years, people searched for a construction. Many believed they would eventually find one. But the difficulty was hidden inside the number π .

What Kind of Numbers Can Geometry Create?

This is where algebra enters. A compass and straightedge construction does not allow arbitrary numbers. Starting from a unit length, every construction produces numbers obtained through repeated operations: +, -, $\times$, /, and $\sqrt{}$.

For example, starting with 1, you can create 2 because $1+1 = 2$. You can create $\sqrt{2}$ by constructing a right triangle. Using the Pythagorean theorem:

$$1^2 + 1^2 = c^2$$

Therefore:

$$c^2 = 2$$

$$c = \sqrt{2}$$

Every number created this way is called an algebraic number. An algebraic number is a number that satisfies a polynomial equation with integer coefficients:

$$a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0 = 0$$

For example, $x^2 - 2 = 0$ has the solution $x = \sqrt{2}$. Therefore, $\sqrt{2}$ is algebraic.



Numbers Beyond Equations: Transcendental Numbers

Some numbers do not satisfy any polynomial equation with integer coefficients. They are called transcendental numbers. They transcend algebra; they escape every equation. Two famous examples are π and e .

The discovery that π was transcendental changed mathematics forever. In 1882, Ferdinand von Lindemann proved that π is transcendental. This meant that π was not merely complicated. It belonged to a completely different category of numbers. It could never be the solution of a polynomial equation with integer coefficients.

Since compass-and-straightedge constructions can only create algebraic numbers, the required length cannot be constructed. Therefore, squaring the circle is impossible. Not difficult. Not undiscovered. Impossible. The universe of geometric constructions simply does not contain the required number.

The Deeper Lesson: A Maze With No Exit

Imagine entering a maze. You walk through corridors searching for an exit. You try turning left, turning right, climbing walls, and walking backwards. After a while someone tells you "There is no exit." That feels different from saying "You have not searched enough."

The first statement changes the nature of the problem. The question is no longer: How do we escape? It becomes: Why was escape impossible?

This is what impossibility proofs do. They do not stop exploration. They redirect it.

Doubling the Cube: The Cube That Refused to Grow

The second famous impossible problem begins with a legend. A plague struck Athens. According to the story, the citizens asked the oracle at Delphi how to stop it. The answer was double the size of the altar. The altar was a cube. The Greeks assumed this meant construct a cube with twice the volume. Simple enough.

A cube with side length a has a volume: $V = a^3$

A doubled cube requires: $2V = 2a^3$

If the new side length is x : $x^3 = 2a^3$



Dividing by a^3 :

$$\left(\frac{x}{a}\right)^3 = 2$$

Therefore:

$$\frac{x}{a} = \sqrt[3]{2}$$

So the problem requires constructing $\sqrt[3]{2}$. Can a compass and a straightedge create this number? The answer is no.

Why Cube Roots Create a Barrier

The number $\sqrt[3]{2}$ is algebraic because it satisfies $x^3 - 2 = 0$. Its algebraic degree is 3. But compass-and-straightedge constructions have a special restriction. Each time a square root is introduced, the degree of the number can multiply by two. This means constructible numbers have algebraic degrees that are powers of two: 1, 2, 4, 8, 16...

The number 3 is not a power of two. Therefore, $\sqrt[3]{2}$ cannot be constructed. The cube cannot be doubled.

Trisecting an Angle: The Problem of Perfect Division

The final famous problem asked: Given any angle, divide it into exactly three equal parts. For some special angles, this is easy. A 90° angle can become $30^\circ + 30^\circ + 30^\circ$, and a 60° angle can become $20^\circ + 20^\circ + 20^\circ$. The problem is: Can we do it for every possible angle? The answer is no. The difficulty again comes from algebra.

Suppose we want to trisect a 60° angle. That means constructing 20° . Now, we know:

$$\cos(60^\circ) = \frac{1}{2}$$

Using the triple-angle identity:

$$\cos(3\theta) = 4\cos^3\theta - 3\cos\theta$$

Let $x = \cos(20^\circ)$. Then

$$4x^3 - 3x = \frac{1}{2}$$

Multiplying by 2:

$$8x^3 - 6x - 1 = 0$$



The construction problem has become a cubic equation. And cubic equations of this type cannot generally be solved through compass-and-straightedge methods. The problem is not the angle. The problem is the algebra hiding underneath.

The First Great Lesson of Impossibility

The three ancient problems reveal a deep principle: a problem may appear geometric, but underneath, it may actually be algebraic. A drawing may secretly ask:

- Can this number exist?
- Can this equation have a solution?
- Does this object belong to the mathematical universe we are working in?

Mathematics often hides one language inside another. Geometry becomes algebra. Algebra becomes logic. Logic becomes philosophy. Eventually mathematics begins asking: Are there questions whose answers are permanently unreachable?

The Dream of a Perfect Mathematical Universe

The ancient Greeks discovered limits caused by their tools. However, mathematics eventually encountered a much deeper question: Are there limits built into mathematics itself? At the beginning of the twentieth century, many mathematicians believed the answer was no. They hoped to build a complete foundation for all of mathematics.

The idea was simple: If mathematics is a logical system, then perhaps we can write down a complete collection of basic rules and assumptions from which every mathematical truth can be derived. Imagine mathematics as a giant building. The foundations would be a small collection of basic principles called axioms. From those axioms, mathematicians would construct everything else: arithmetic, geometry, algebra, calculus, and number theory. Every theorem would be like another brick in the building. The dream was that every brick could be traced back to the foundation. Nothing would be uncertain. Nothing would be missing. Every true mathematical statement would eventually be provable.



THE DREAM OF COMPLETENESS

This vision became known as formalism. One of its strongest supporters was the German mathematician David Hilbert. Hilbert believed mathematics could be placed on an absolutely secure logical foundation. His ambition was that mathematics should satisfy two conditions:

1. Completeness: Every true statement should be provable.
2. Consistency: The system should never produce contradictions.

The dream was that mathematics could become a perfect logical machine. A question would be asked. The axioms would be applied. A proof would emerge. Every truth would have a path back to the foundations. But mathematics was about to discover something astonishing: Some truths have no such path.

THE STRANGE POWER OF SELF-REFERENCE

The key idea behind the coming revolution was surprisingly simple: A system can sometimes talk about itself. This happens in ordinary language. Consider the sentence: "This sentence is false." What happens? If the sentence is true, then what it says must happen, meaning it is false. But if it is false, then what it says is not happening, meaning it is true.

The statement creates a loop. It refers to itself. This is called self-reference. For centuries, philosophers studied examples like this. However, mathematicians began to realise something deeper: Self-reference was not merely a philosophical curiosity. It could become a mathematical tool. A system powerful enough to describe itself might also be capable of describing its own limitations. This idea would become central to the work of Kurt Gödel.

GÖDEL'S INCOMPLETENESS THEOREM

In 1931, Kurt Gödel published a result that completely changed mathematics. His theorem showed that mathematics cannot be completely captured by a single formal system. Gödel proved: Any sufficiently powerful mathematical system will contain statements that are true but cannot be proven within that system. This was called the incompleteness theorem. To understand why this is so remarkable, we first need to understand what Gödel did.



TURNING MATHEMATICS INTO NUMBERS

Gödel's genius was realising that mathematics could describe itself using mathematics. He created a method called Gödel numbering. The basic idea was: Assign every mathematical symbol a number. For example, imagine:

$$+ \rightarrow 5, = \rightarrow 8, 0 \rightarrow 12, 1 \rightarrow 13$$

A mathematical statement could then be converted into a sequence of numbers. A proof could become a number. A formula could become a number. A statement about mathematics could itself become a mathematical object. Mathematics had gained the ability to look into a mirror.

THE IMPOSSIBLE STATEMENT

Gödel then constructed a mathematical statement that essentially said: "This statement cannot be proven." At first, this sounds like a contradiction. It resembles the famous liar paradox: "This sentence is false." However, Gödel's statement is different. The liar paradox creates a contradiction because it talks about truth and falsity. Gödel's statement talks about provability.

Now consider what happens. Suppose the statement is false. If the statement is false, then what it claims is not true. Therefore, the statement actually can be proven. But if the statement can be proven, then the mathematical system proves a false statement. That would mean the system is inconsistent. However, mathematicians wanted systems that were consistent. Therefore, if the system is consistent, Gödel's statement cannot be false. So it must be true.

The conclusion is extraordinary: The statement is true, but it cannot be proven. A mathematical truth exists beyond proof.

WHY GODEL CHANGED MATHEMATICS FOREVER

Before Gödel, many mathematicians imagined mathematics as a machine. The machine would work like this:

$$\textit{Axioms} \rightarrow \textit{Logical Rules} \rightarrow \textit{Every Mathematical True}$$

If we knew the rules and applied them carefully enough, every answer would eventually appear. Gödel showed that this machine could never be complete. No matter how carefully the system is designed, if it is powerful enough to describe ordinary arithmetic, there will always be statements that remain outside its reach. You can expand the system by adding new axioms. But then the expanded system will contain new statements that cannot be proven. You can build a larger mathematical building. But there will always be rooms beyond the blueprint.

This does not mean mathematics is broken. It means mathematics discovered something about its own structure. Mathematics proved a fact about mathematics itself.



AN ANALOGY: THE ULTIMATE INSTRUCTION MANUAL

Imagine someone tries to write the perfect instruction manual. The manual explains every possible rule for every possible situation. It contains a section explaining how to read the manual. It contains another section explaining how to interpret those explanations. Then another section explaining the interpretation of those interpretations. Eventually, the manual becomes part of the system it is describing. Can the manual completely describe itself?

This is the difficulty Gödel uncovered. A system powerful enough to describe itself becomes trapped by self-reference. It cannot completely stand outside itself. A map can describe a landscape. But a map of the entire universe would need to include itself. Where would that map be placed? The moment it includes itself, the problem becomes circular.

FROM MATHEMATICS TO COMPUTERS

Gödel showed that some mathematical truths escape proof. A few years later, another mathematician asked a related question: What about computation? Could we build a machine that solves every possible problem? Could a computer examine any program and determine whether it will eventually finish or continue forever?

This question was practical rather than philosophical. A computer program either completes its task or continues endlessly. Surely, a sufficiently intelligent computer could determine which one occurs. The answer was another impossibility. The mathematician who discovered this was Alan Turing.

THE HALTING PROBLEM: A QUESTION NO COMPUTER CAN ALWAYS ANSWER

Imagine creating a program called the ultimate program checker. Its job is simple: Given any computer program, it determines whether that program eventually stops or runs forever.

For example:

Program A :
count from 1 to 10
stop

The checker says : Stops (Correct)

Program B :
count forever
The checker says : Runs forever (Correct)

Now imagine feeding the checker its own code. Ask: "Will you stop?"



TURING'S CONTRADICTION

Assume there exists a perfect checker called H. The checker takes a program P and returns:

$$H(P) = \begin{cases} \text{stops, if } P \text{ eventually finishes} \\ \text{loops, if } P \text{ runs forever} \end{cases}$$

Now construct another program Q. Program Q does the opposite of whatever H predicts. If $H(Q) = \text{stops}$, then Q runs forever. If $H(Q) = \text{loops}$, then Q stops. In both cases, the checker fails. Therefore, the perfect checker cannot exist.

Turing proved: No algorithm can solve the halting problem for every program. This result became one of the foundations of theoretical computer science.

THE SAME PATTERN APPEARS AGAIN

The ancient Greeks discovered: some geometric constructions cannot exist. Gödel discovered: some mathematical truths cannot be proven. Turing discovered: some computational problems cannot be solved. These are not isolated accidents. They reveal a deep mathematical principle: Every sufficiently powerful system has boundaries. Every mathematical world operates according to rules. And once a system becomes powerful enough, those rules create unavoidable limitations.

THE BEAUTY OF IMPOSSIBILITY

At first, impossibility sounds negative. A problem has no solution. A question has no answer. A goal cannot be achieved. However, mathematics reveals something more subtle. Knowing what cannot be done is itself knowledge. Before the proof that the circle could not be squared, people spent centuries searching for a construction that could never exist. After Gödel, mathematicians stopped searching for a complete formal system that could prove everything. After Turing, computer scientists understood that some problems require different approaches because no universal solution can exist.

Impossibility does not close the door on mathematics. It tells us which doors are real doors and which are painted onto the wall.



THE MATHEMATICS OF PREDICTION

The impossibility results of geometry, logic, and computation all reveal the same idea: A limitation may not come from a lack of intelligence, effort, or technology. Sometimes the barrier is built into the structure of the problem itself. Perhaps the strangest example appears when humans try to predict the future. For centuries, people dreamed of a universe that behaved like a perfect machine. If we knew enough information about the present, perhaps we could calculate everything that would happen next. This leads to one of mathematics' deepest questions: Can a completely deterministic universe still be impossible to predict?

THE UNIVERSE AS A MACHINE

In the eighteenth and nineteenth centuries, many scientists imagined the universe as a giant mechanical system. The famous French mathematician Pierre-Simon Laplace described an imaginary intelligence that became known as Laplace's demon. This was not a real creature. It was a thought experiment.

Imagine an intelligence that knows:

- The exact position of every particle in the universe;
- The exact speed and direction of every particle;
- Every physical law governing their interactions.

Would this intelligence be able to calculate the entire future? According to classical physics, the answer appeared to be yes. The universe would be like an enormous clock. If you knew the position of every gear and every spring, you could predict every movement forever. This idea is called determinism. A deterministic system is a system in which the current state completely determines the next state:

$$\textit{Present State} \rightarrow \textit{Future State}$$

There is no randomness. No mystery. No uncertainty. But mathematics eventually discovered something unexpected: Determinism does not guarantee predictability.

A TINY CHANGE THAT BECOMES A HUGE DIFFERENCE

the height, gravity, air resistance, and initial velocity, then we can calculate where the ball will land.

Now imagine a much more complicated system: the atmosphere. A tiny change in atmospheric conditions might be 0.000001 degrees of temperature difference. At first, this seems meaningless. However, the atmosphere is a connected system. A small difference can influence air currents, clouds, sunlight absorption, temperature, pressure, and winds. The tiny difference grows.

Eventually, two almost identical starting conditions create completely different outcomes. The system is still following exact equations. Nothing random has happened. The problem is that the future becomes extremely sensitive to the present. This phenomenon is known as sensitive dependence on initial conditions. More commonly, it is called the butterfly effect.



THE BUTTERFLY EFFECT: CHAOS HIDDEN INSIDE SIMPLE EQUATIONS

The phrase “butterfly effect” comes from the idea that a butterfly flapping its wings could eventually influence a distant storm. This does not mean that a single butterfly literally creates a hurricane. The mathematical idea is deeper: A tiny change in the initial conditions of a complex system can eventually produce a dramatically different result. The surprising part is that chaos does not require complicated rules. Some of the simplest equations in mathematics can create chaotic behaviour.

THE LOGISTIC MAP

One famous example is the logistic map:

$$x_{n+1} = rx_n(1 - x_n)$$

At first glance, this equation seems harmless. It produces a sequence of numbers. Choose an initial value and a parameter r . Then repeatedly calculate:

$$x_1 = rx_0(1 - x_0)$$

Followed by:

$$x_2 = rx_1(1 - x_1)$$

and so on. The equation is simple enough to write on one line. Yet for certain values of r , the behaviour becomes incredibly complicated. The sequence may settle into a stable pattern, oscillate between values, or become completely chaotic. The equation is deterministic. There is no randomness. Yet long-term prediction becomes impossible.

x_0

WHY PREDICTION FAILS

The reason chaos creates impossibility is not because the equations are unknown. The equations may be perfectly known. The problem is measurement. Suppose we measure a temperature as 20.5°C. The actual temperature might be 20.503847291°C. The difference is tiny and for ordinary situations, it does not matter. But in a chaotic system, that tiny difference can grow exponentially.



Mathematically, the separation between two nearby states can behave approximately as:

$$d(t) = d(0)e^{\lambda t}$$

where:

- $d(0)$ is the initial difference;
- $d(t)$ is the difference after time ;
- λ is the Lyapunov exponent.

If $\lambda > 0$ then small errors grow exponentially. The future becomes impossible to calculate accurately beyond a certain point.

THE STRANGE IDEA OF A PREDICTABLE BUT UNKNOWABLE SYSTEM

This leads to a subtle distinction. A chaotic system is not random. Randomness means there is no determined pattern. Chaos means there is a pattern, but the pattern is too sensitive for practical prediction.

Imagine a perfectly balanced pencil standing upright on its tip. The laws of physics determine what happens. However, the slightest disturbance—a vibration, a tiny air current, a microscopic irregularity—can decide whether it falls left or right. The laws are not uncertain. Our knowledge of the starting conditions is. The universe may be deterministic, but our ability to predict it is limited.

THE THREE-BODY PROBLEM: WHEN GRAVITY BECOMES IMPOSSIBLE TO CALCULATE

One of the most famous examples comes from astronomy. The motion of two objects under gravity is beautifully predictable. For example, Earth orbiting the Sun or the Moon orbiting Earth. Mathematicians can derive equations describing their motion. However, add a third object and everything changes. This is called the three-body problem.

Imagine three planets interacting through gravity. Each planet pulls on the others. Planet A affects planet B. Planet B affects planet C. Planet C affects planet A. The interactions continuously change. The equations become extraordinarily complicated. For centuries, mathematicians searched for a general formula that would predict the motion of three bodies forever. Eventually, Henri Poincaré showed that even deterministic gravitational systems could contain chaotic behaviour. This discovery became one of the foundations of modern chaos theory.

THE LIMITS OF WEATHER PREDICTION

Weather is perhaps the most familiar example of mathematical unpredictability. Weather models are based on enormous systems of equations describing temperature, pressure, humidity, wind, and ocean currents. Modern computers solve billions of calculations to produce forecasts. These predictions are incredibly useful. However, they have a limit. A forecast several days ahead may be reliable. A forecast several months ahead becomes much harder. This is not simply because computers are not powerful enough. Even with perfect computers, chaotic systems create a fundamental limit. A computer cannot calculate information that does not exist. If the atmosphere's initial state is not known with infinite precision, eventually the uncertainty becomes too large. The problem is not computational weakness. The problem is mathematical structure.



PROBABILITY; WHEN CERTAINTY DISAPPEARS

Another major example of mathematical limitation comes from probability. Before probability theory developed, people often viewed uncertainty as simply ignorance. If we did not know what would happen, perhaps that was only because we lacked enough information. Probability revealed a deeper possibility: Some systems are naturally described through uncertainty.

THE MATHEMATICS OF RANDOMNESS

A common misconception is that randomness means "without mathematics." The opposite is true. Randomness has some of the deepest mathematics behind it. A single coin toss seems unpredictable: H or T. However, thousands of coin tosses follow mathematical patterns. The probability of exactly k heads in n tosses is:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

For a fair coin, $p = 0.5$. The individual event is uncertain. The overall pattern is predictable. This is one of mathematics' most beautiful ideas: Individual outcomes can be impossible to predict, while collections of outcomes can obey extremely precise laws.

WHEN IMPOSSIBLE BECOMES POSSIBLE: EXPANDING THE MATHEMATICAL

UNIVERSE One of the most fascinating lessons from impossible problems is that sometimes the answer is not to force the old system to change. Sometimes the answer is to create a larger system. Mathematics repeatedly expands itself when it encounters something that appears impossible. The impossible object is often not impossible everywhere. It is only impossible under the original rules. This idea appears throughout the history of mathematics.

THE IMPOSSIBLE SQUARE ROOT

Consider the equation:

$$x^2 = -1$$

At first, this appears impossible. No real number satisfies it. Positive numbers square to positive values ($2^2 = 4$), and negative numbers also square to positive values ($(-2)^2 = 4$). Therefore, $x^2 = -1$ has no solution among the real numbers. For centuries, mathematicians considered this problem deeply uncomfortable. The square root of a negative number seemed meaningless.

But instead of rejecting the problem, mathematicians expanded the number system. They created a new number:

$$i = \sqrt{-1}$$

with the property:

$$i^2 = -1$$



This created the complex numbers. The complex numbers were once considered artificial. Today they are essential. They appear in electrical engineering, quantum mechanics, signal processing, computer graphics, and control theory. The impossible became useful. The problem was not that mathematics was wrong. The problem was that mathematics was incomplete.

NON-EUCLIDEAN GEOMETRY: WHEN PARALLEL LINES MEET

A similar revolution occurred in geometry. For thousands of years, Euclid's geometry was considered the only possible geometry. One of its famous assumptions was the parallel postulate: "Given a line and a point not on that line, there is exactly one parallel line passing through the point." Mathematicians attempted for centuries to prove that this statement followed from the other geometric axioms. They failed.

Eventually, they discovered something extraordinary: The parallel postulate was not necessary. Other geometries could exist. In curved spaces, parallel lines behave differently. On a sphere, for example, the closest equivalent of a straight line is a great circle. Two such paths that appear parallel on a flat map may eventually meet. A triangle drawn on a sphere can have angles adding to more than 180° . For example, $90^\circ + 90^\circ + 90^\circ = 270^\circ$. This is impossible in Euclidean geometry, but perfectly valid in spherical geometry.

THE BIRTH OF NEW MATHEMATICIANS

The discovery of non-Euclidean geometry changed mathematics forever. The lesson was profound: A statement can be impossible inside one mathematical universe and completely natural inside another.

The same idea appears again and again:

- Real numbers cannot contain $\sqrt{2}$, so mathematics creates complex numbers.
- Euclidean geometry cannot describe curved spaces, so mathematics creates non-Euclidean geometry.
- Classical logic cannot prove every truth, so mathematics studies stronger logical systems.
- Computers cannot solve every problem, so computer science studies the boundaries of computation.

Mathematics grows by discovering its limitations.

WHY MATHEMATICIANS LOVE IMPOSSIBLE PROBLEMS

At first glance, an impossible problem seems disappointing. Why spend centuries studying something that cannot be achieved? The answer is that impossible problems reveal hidden structures.

A solved problem tells us: "Here is the answer." An impossible problem tells us: "Here is why no answer could ever exist." The second statement is often deeper. A solution gives information about one object. An impossibility proof gives information about an entire universe of objects. For example, proving that one equation has no solution tells us something about that equation. Proving that no equation of a certain type can be solved tells us something about all equations of that type. The scope becomes much larger.



THE UNEXPECTED CREATIVITY OF FAILURE

Human creativity is often associated with success. However, mathematics shows that failure can be equally powerful. The Greeks failed to square the circle, but their failure led to a deeper understanding of numbers. Mathematicians failed to create a complete logical system, but that failure led to modern logic and computer science. Scientists failed to predict chaotic systems perfectly, but that failure led to a new understanding of complexity. The boundary itself became the discovery.

A NEW DEFINITION OF PROGRESS

In many areas of life, progress means overcoming limitations. Mathematics reveals another form of progress: Understanding limitations. Knowing that something cannot be done is not the end of knowledge; it is often the beginning. Before an impossibility proof, humanity wastes effort searching for a nonexistent solution. After an impossibility proof, humanity can ask better questions. Instead of: "How do we do this?" we ask: "Why can this not be done?" That question is often where the deepest mathematics begins.

THE FINAL PARADOX

Mathematics is the study of possibility. But some of its greatest discoveries are discoveries of impossibility. This creates a beautiful paradox. The subject that allows humans to calculate, construct, and predict also reveals where calculation, construction, and prediction must stop.

- The ancient Greeks discovered: Some shapes cannot be constructed.
- Algebra discovered: Some equations cannot be solved.
- Gödel discovered: Some truths cannot be proven.
- Turing discovered: Some problems cannot be computed.
- Chaos theory discovered: Some futures cannot be predicted.

CONCLUSION: THE MAP OF POSSIBLE

Mathematics is often described as the search for truth, but truth does not always appear as a solution. Sometimes truth appears as a boundary. A mathematician is not only someone who discovers new paths. A mathematician is also someone who carefully maps where paths cannot exist.

The greatest mathematical discoveries are sometimes not the construction of new objects. They are the proof that certain objects can never exist. This does not make mathematics smaller. It makes mathematics deeper. Every impossibility proof tells us something fundamental about the world of ideas. It tells us where the universe of mathematics ends, and perhaps that is one of the most beautiful ideas of all: The limits of mathematics are not failures of human imagination; they are discoveries about the structure of reality itself.

To prove that something is impossible is not to close a door. It is to understand why that door was never there. The most profound answers in mathematics are not always the ones that say "yes". Sometimes the most beautiful answer is a perfectly proven "no."

Next-Generation Aircraft and Avionic Systems





ABSTRACT

For over two decades, the aviation sector has sought lightweight, passive methods of recovering energy from aerodynamic loads and reducing carbon emissions. Existing research includes limited experimentation with the placement of piezoelectric (PZT) sensors on aircraft wings, and the mediating role of angle of attack (AoA) has not been widely investigated. This study compared the electrical output of PZT ceramic disk sensors mounted at three wing locations, 50%, 75%, and 100% of span measured from root to tip, to capture flow-induced vibrations. A three-dimensional (3D)-printed wing model with piezoelectric sensors attached at each position was tested in a wind tunnel across three AoA settings, -10° , 0° , and 10° , to vary vortex strength and vibration intensity. Voltage production was recorded using a multimeter, and drag was recorded from the wind tunnel display. Wing geometry, sensor orientation, and sensor type were held constant to reduce variability. Two-sided Welch t-tests ($\alpha = 0.05$) showed that sensors mounted at the 50% placement produced greater electrical output than the 75% placement ($p = 0.0018$) and the 100% placement ($p < 0.001$) at 0° AoA, although this placement also produced a statistically significant drag penalty relative to the baseline and other placements ($p < 0.001$). Placement selection therefore depends on mission priorities: maximising power or minimising drag. This proof-of-concept study demonstrated that targeted piezoelectric placement can recover useful energy for onboard sensors and low-power electronics in avionics and, with further optimisation, could be adapted for other fields such as the automotive and marine industries.





INTRODUCTION

Aircraft are indispensable to modern transport, commerce, and reconnaissance, supporting activities ranging from passenger travel to global trade. According to the Federal Aviation Administration, over sixteen million flights are scheduled annually, and this figure is projected to double [1]. Although air travel is far more efficient than it was in 1990, when a passenger-kilometre required 2.9 megajoules (MJ) of energy, by 2019 it still required 1.3 MJ per passenger-kilometre, totalling roughly one billion tonnes of carbon dioxide (CO₂) annually [2]. Between 2000 and 2019, emissions from aviation grew faster than those from rail, road, or shipping, highlighting the sector's increasing environmental impact [3]. As the aviation sector and the energy demands of onboard avionics and sensors continue to grow, engineers face increasing pressure to develop reliable, sustainable power solutions.

Wingtip vortices form as air spills from the high-pressure region beneath a wing into the low-pressure region above it. Researchers have explored aerodynamic refinements and energy-harvesting concepts aimed at reducing wake losses and recovering energy from such structures [4]. Although traditionally treated only as a source of aerodynamic inefficiency to be reduced through devices such as winglets, these vortices also concentrate recoverable kinetic energy, presenting an opportunity to convert mechanical stress into small-scale electrical power [5]. One method of harvesting this energy is vortex-induced vibration (VIV) harvesting, in which airflow oscillations create structural vibrations that can be converted into electricity using piezoelectric materials [6]. Piezoelectric elements are useful in aerospace contexts because they are lightweight, compact, and able to generate electricity directly from mechanical strain without additional moving parts [6]. Proof-of-concept studies using such materials have demonstrated power outputs in the microwatt-to-milliwatt range, but scaling this technology for practical aircraft applications requires targeted experiments that weigh aerodynamic penalties against energy gains [7].

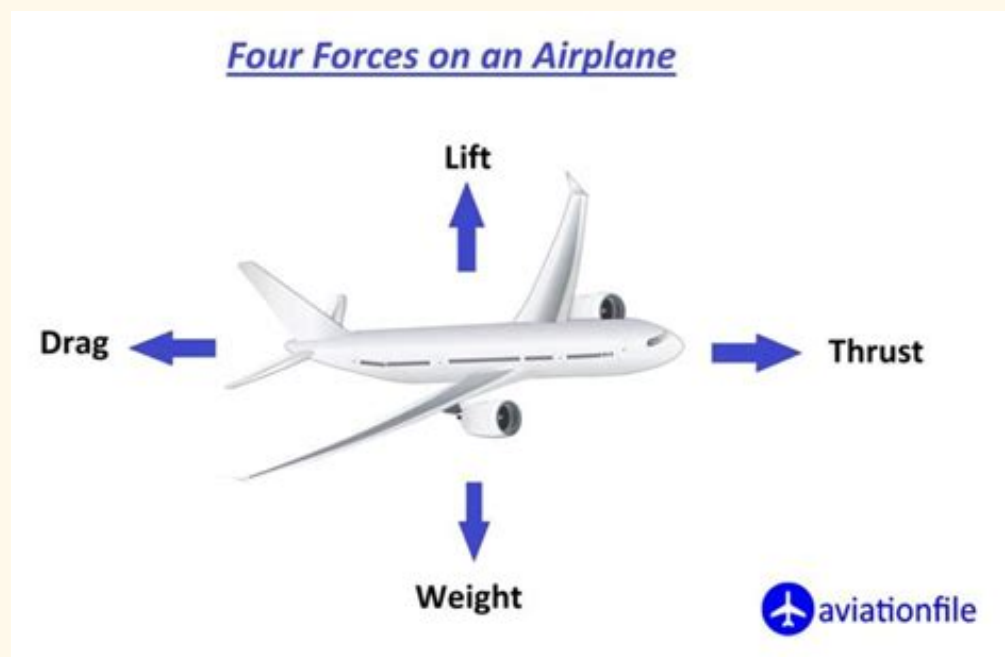
Despite decades of research aimed at minimising induced drag, the underutilisation of wingtip vortex energy for powering avionics and onboard electronics remains a gap in current understanding. Existing literature includes limited experimentation on the placement of piezoelectric sensors along a wing and on the mediating role of angle of attack. This study tested piezoelectric ceramic sensors at three winglet locations across a range of angles of attack, measuring electrical output and drag change against a baseline winglet without a sensor attached. By assessing whether piezoelectric harvesters can reclaim a fraction of vortex energy without introducing significant aerodynamic penalties, this research aims to expand the design space for sustainable avionics systems and to provide a pathway toward self-sustaining, lightweight power solutions for future aircraft.



The engineering goal of this project was to achieve at least one milliwatt of output per ceramic, printed-circuit-board (PCB)-mounted piezoelectric sensor during cruise-like vortex excitation, while keeping added mass below three per cent of winglet mass and drag increase under five per cent. The hypothesis was that if piezoelectric sensors were attached to a winglet at different locations and the winglet was tested at multiple angles of attack, the sensor placed at the 50% span location under a 10° angle of attack would produce the highest energy output while causing a minimal change in drag.

LITERATURE REVIEW

Flight depends on the interaction of four primary forces: lift, weight, thrust, and drag [8]. Lift allows an aircraft to overcome gravity and remain airborne. Aircraft wings are engineered with an aerofoil cross-section, in which the top surface is curved and the bottom surface is flatter [9]. As air flows over the wing, the curved surface accelerates the air above while the air beneath moves more slowly, creating a pressure difference that generates an upward force through Bernoulli's principle, in which faster airflow corresponds to lower pressure [9]. The aerofoil's camber, thickness, and angle of attack determine the lift coefficient (C_l), which quantifies how effectively a wing converts oncoming airflow into lift [10]. As AoA increases, the wing accelerates flow over the upper surface and deflects more air downward, so C_l rises and lift increases until flow separation causes a stall [10]. Generating lift also produces induced drag, a by-product of the pressure differences that push air around the wingtips [11]. This induced downwash tilts the lift vector rearward, producing induced drag that increases especially at lower speeds or high angles of attack, since generating extra lift requires redirecting more airflow into the wake [11].





One of the most influential refinements to wing design has been the winglet, a small vertical or canted extension added to a wingtip to alter wingtip aerodynamics and improve efficiency [4]. Winglets reshape airflow at the wingtip, where pressure differences between the upper and lower wing surfaces cause high-pressure air beneath the wing to spill over into the low-pressure region above, forming wingtip vortices [12]. These vortices are spiralling, trailing airflows that generate downwash behind the wing, tilting the effective relative wind downward and forcing part of the wing's lift vector to tilt backward, producing induced drag that the engines must overcome [13]. Stronger vortices increase induced drag and reduce aerodynamic efficiency [13]. Winglets disrupt this circulation by redirecting tip airflow upward or outward, weakening vortex strength and reducing downwash, though they do not eliminate wingtip vortices entirely; they weaken and reorganise tip circulation while leaving residual vortex strength that continues to influence wake behaviour [12]. Winglets are most beneficial during cruise at high altitude and during phases requiring high lift, such as take-off or climb, when induced drag is largest, which explains their widespread adoption across passenger, cargo, and business aircraft [14]. Modern aircraft from both Airbus and Boeing widely incorporate winglet variants, including sharklets and raked tips [15].

The piezoelectric effect is a physical phenomenon in which certain noncentrosymmetric crystals and engineered materials generate electric charge when mechanically stressed, and conversely deform when an electric field is applied [16]. Mechanical strain moves positive and negative ions inside the material's unit cell, the smallest repeating structural block; each shifted unit cell becomes a tiny electric dipole, and when many dipoles align, a macroscopic polarisation forms, producing the measurable voltage that appears across the element under stress [17]. Manufacturers increase and lock in this response through poling, a process that uses a strong electric field to align internal domains so their dipoles do not cancel [18]. This coupling is reciprocal: applied mechanical stress produces a measurable charge and voltage, while an applied electric field produces a small mechanical strain, which underpins both sensing and energy-conversion applications [18]. Relevant piezoelectric materials include piezoceramics such as lead-zirconate-titanate (PZT) and piezopolymers such as polyvinylidene fluoride (PVDF); these families differ substantially in electromechanical coefficients and mechanical properties [19]. Typical PZT ceramics exhibit d_{33} values on the order of hundreds of picocoulombs per newton, while PVDF films show d_{33} values on the order of tens of picocoulombs per newton, explaining why ceramics deliver higher power density while polymers offer greater flexibility [19]. Device-level power extraction depends on concentrating strain at the harvester, through choices such as multilayer stacks, bimorphs, cymbal transducers, or flexible films, so that the piezoelectric element operates near the dominant vibration frequency [18]. Vibration sensors and accelerometers provide an everyday precedent for converting small motions into electrical signals using this fast, local piezoelectric response [20].



Vortex-induced vibration harvesting has been explored across laboratory configurations, including cylinders, cantilevers, and oscillating foils, as well as preliminary wing-scale experiments, consistently demonstrating that energy capture from VIV is physically achievable [5], [21]. However, power output is generally modest, typically in the micro- to milliwatt range per harvester, unless the geometry and modal properties of the piezoelectric device are carefully matched to the natural frequency of the structure and the excitation frequency of the vortices [21]. Feasibility analyses emphasise several engineering trade-offs when integrating piezoelectric strips onto winglets: designers must balance electromechanical coupling against added drag and structural mass, while ensuring the harvester does not compromise the aeroelastic stability of the winglet under variable flight conditions [22]. Existing research combining wind-tunnel experiments on realistic winglet geometries with coupled electromechanical optimisation remains limited [6].

Piezoelectric harvesters integrated with winglets offer potential benefits alongside limitations. Winglets alone have demonstrated typical cruise fuel burn reductions of three to six per cent, equivalent to tens of gallons per flight hour [23]. Piezoelectric harvesters, by contrast, generally produce nanowatts to milliwatts per device under realistic flight vibration or vortex conditions, sufficient for powering wireless sensors or telemetry, but orders of magnitude below propulsion needs [18]. Key limitations include fatigue of piezoelectric stacks and polymer films under cyclic aerodynamic loading, added mass and structural complexity, low energy-conversion efficiency in aeroelastic environments, and high per-watt costs compared with conventional power sources; these factors can lengthen payback periods or negate fuel savings unless harvesters are lightweight, durable, and aerodynamically neutral [22]. For harvesters to be worthwhile, they must deliver measurable onboard value in terms of sensor power and reduced battery replacement, adhere to strict mass and drag constraints, and demonstrate lifecycle savings that justify installation and certification costs [24]. The most realistic near-term application is therefore hybrid: using piezoelectric harvesters to enable self-powered health monitoring, reduce maintenance logistics, and marginally extend the endurance of uncrewed aerial vehicles, while longer-term materials advances could raise power density and lower cost [25].



METHODS

This study was conducted indoors at Shrewsbury Senior High School. All soldering was carried out under adult supervision while wearing impact-resistant safety goggles and heat-resistant gloves, and all wind-tunnel operation was conducted under adult supervision.

INDEPENDENT VARIABLES

Two independent variables were tested: the location of the piezoelectric sensor on the wing model, and the angle of attack (AoA) of the wing. Sensor location was set to 50%, 75%, and 100% of wing span, measured from the root of the wing model to the wingtip, progressively moving the PCB-mounted piezoelectric ceramic sensor closer to the wingtip. For each sensor location, AoA was set to 10°, 0°, and -10° to produce varying vortex strengths, representing angles typically observed during take-off, cruise, and landing.

DEPENDENT VARIABLES

The dependent variables were the aerodynamic drag produced by the wing model, measured in volts (V) as output by the wind-tunnel's drag sensor, and the electrical output generated by the mounted piezoelectric sensor, also recorded in volts. As the drag sensor produces a voltage proportional to drag force and a calibration to newtons was not available for this apparatus, drag results are reported in volts and compared relatively across conditions rather than as absolute force values.

CONTROL VARIABLES

The control condition was an unmodified wing model without a piezoelectric sensor attached, tested at the same three angles of attack. Drag produced by the unmodified model was compared with drag produced by the modified models to determine the aerodynamic impact of adding the sensor. Energy output for the unmodified wing was taken to be zero, as is the case for all unmodified wings.

MATERIALS AND FABRICATION

A wing model based on a KC-135 aerofoil-series winglet design was produced using computer-aided design (CAD) software and exported as a stereolithography (STL) file. The model was sliced with a layer height of 0.20 mm, 20% gyroid infill, and 1.2 mm wall thickness, and printed in polylactic acid (PLA) filament on a Prusa i3 three-dimensional (3D) printer at a nozzle temperature of 200°C and a bed temperature of 60°C. The model was scaled to 10% of the original file size to satisfy the wind tunnel's 10% blockage rule, calculated as $\text{blockage ratio (\%)} = (\text{cross-sectional area of wind tunnel} \div \text{frontal area of model}) \times 100$. Two identical wing models were printed, one for testing and one as a spare; any model deviating from the intended dimensions by more than 2% was reprinted.

A piezoelectric ceramic disk sensor was mounted at each test location using 3M VHB removable indoor mounting tape and connected to two lengths of 22 American Wire Gauge (AWG) wire by soldering, with joints insulated using heat-shrink tubing and the wire secured to the wing using non-conductive tape. Wire crimps were fitted to the exposed wire ends to allow the sensor complex to be moved between the 50%, 75%, and 100% span locations across trials, and the sensor and wing assembly were gently vibrated after mounting to confirm the connection could withstand impact and vibration without detaching. A mounting attachment compatible with the wind tunnel's holding fixture was affixed to the underside of the winglet using hot glue and allowed to cure for five minutes before testing.

EXPERIMENTAL SET-UP

Figure 1 shows the wind tunnel used for testing (LJ Technologies Aerostream Aerodynamics 2000 Scantek), including the dimensions of the holding chamber and mounting fixture.

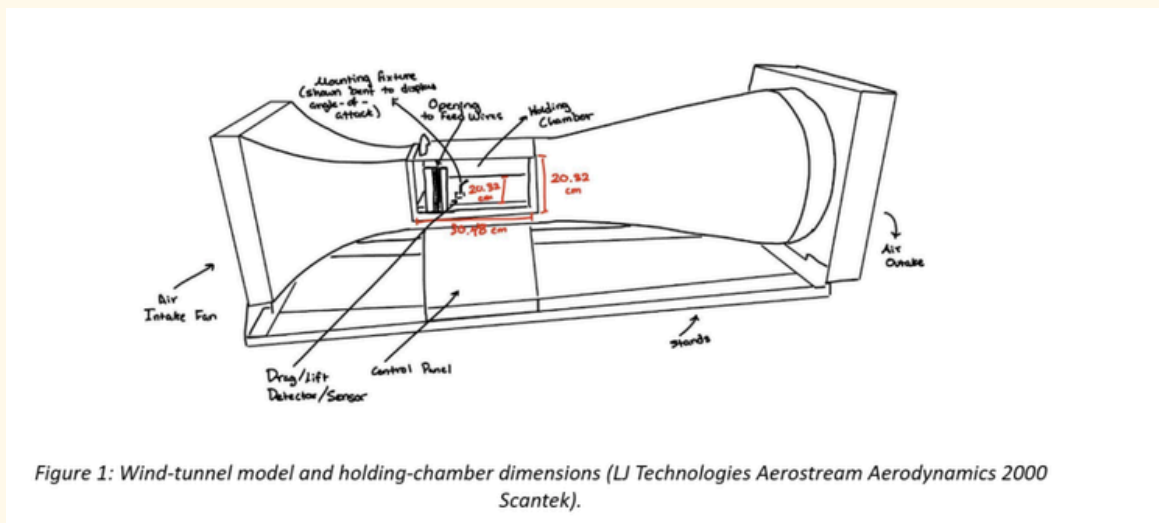


Figure 1: Wind-tunnel model and holding-chamber dimensions (LJ Technologies Aerostream Aerodynamics 2000 Scantek).

Figure 2 shows how the wing model was situated within the wind tunnel during testing, illustrated here with the baseline (unmodified) wing model attached to the mount at a 0° angle of attack.

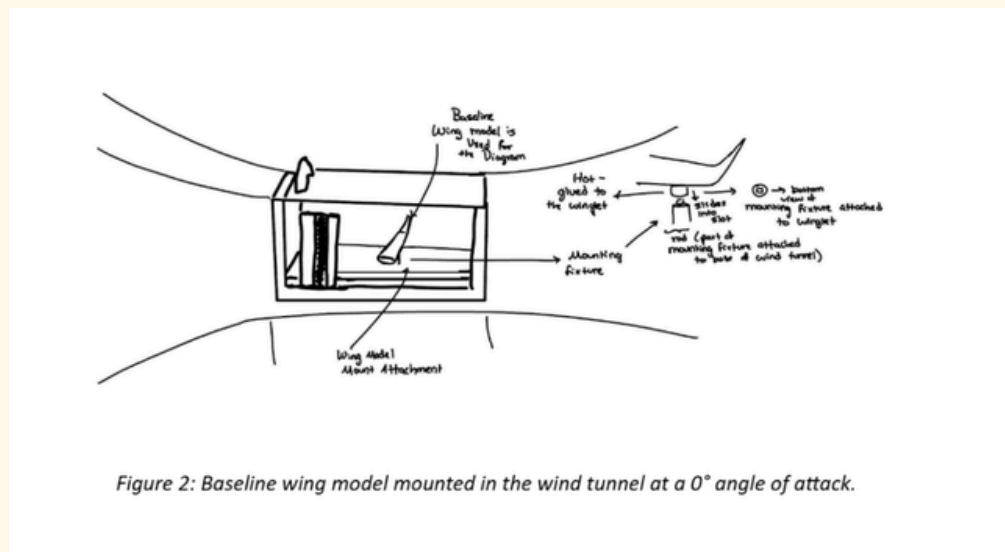


Figure 2: Baseline wing model mounted in the wind tunnel at a 0° angle of attack.



Figure 3 shows the wing model geometry together with the three piezoelectric sensor placements tested (50%, 75%, and 100% of span) alongside the unmodified baseline geometry.

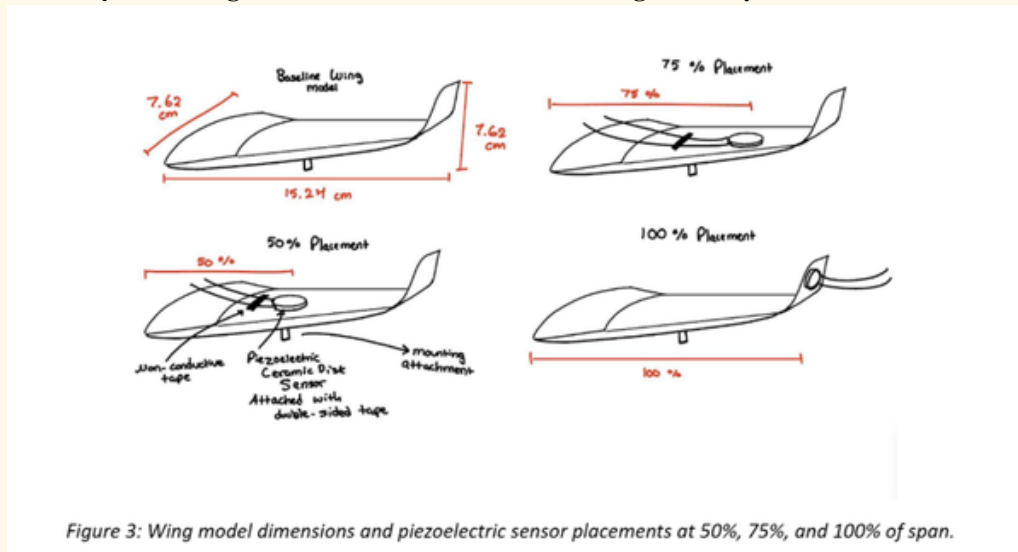
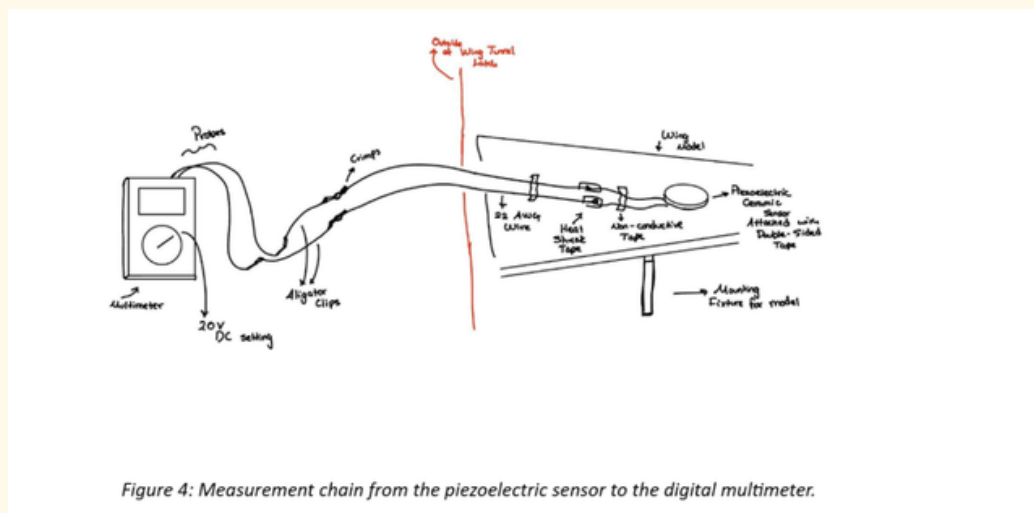


Figure 4 illustrates the full measurement chain, from the piezoelectric ceramic sensor through the soldered leads and crimped connections to the multimeter probes.



TESTING PROCEDURE

The wing model was installed on the mounting stick inside the wind tunnel's holding chamber and set to the required angle of attack, verified to within $\pm 0.5^\circ$ using a protractor before each trial. Sensor leads were passed through the tunnel's accessory pass-through and connected to a digital multimeter (± 0.01 V) using alligator clips, with the multimeter set to the 20 V direct current (DC) range. For each trial, the drag offset was zeroed, the fan was switched on and allowed to stabilise to within 2 mph of the target airspeed, and both the drag reading (in volts, from the wind-tunnel display) and the voltage output (from the multimeter) were recorded before the fan was switched off. Ten trials were completed at each combination of sensor placement (50%, 75%, 100%, and baseline) and angle of attack (10° , 0° , and -10°), giving a total of 120 trials.

STATISTICAL ANALYSIS



Two-sided Welch t-tests were applied to the drag and energy data to determine whether differences between sensor placements, and between each placement and the baseline, were statistically significant. Welch's t-test was selected because it does not assume equal variances between groups. Comparisons were made independently for each angle of attack across the ten trials recorded per condition. A significance threshold of $\alpha = 0.05$ was used to interpret results; p-values below 0.001 are reported as $p < 0.001$, and all other p-values are reported to four decimal places. For each pairwise comparison, the null hypothesis was that the mean measurement, drag or energy, was equal between the two conditions being compared, and the alternative hypothesis was that the means differed. Results

All measurements are reported in volts (V). This unit was used for both the drag produced by each wing configuration and the electrical energy produced by the piezoelectric sensor at each placement, under three angles of attack (10° , 0° , and -10°). As the wind tunnel's drag sensor outputs a voltage proportional to drag force, and calibration to newtons was not available, drag results are reported in volts and compared relatively across conditions.

DATA TABLES

Trial	Baseline	50% Placement	75% Placement	100% Placement
1	5.5	6.1	6	5.1
2	5.6	6.3	5.9	5.7
3	5.1	6.1	5.8	5.5
4	5.3	6	5.9	5.5
5	5.4	6.4	5.9	5.6
6	5.2	6.2	5.7	5.4
7	5.6	5.9	6	5.8
8	5.5	6.1	5.8	5.6
9	5.8	5.9	6	5.5
10	5.3	5.8	5.9	5.4
Mean	5.4	6.1	5.9	5.5
SD	0.2	0.2	0.1	0.2

Table 1: Drag Produced (V) from Placement Types at a 10° Angle of Attack



Placement	Baseline	50%	75%	100%
Baseline	1	< 0.001	< 0.001	0.3862
50%	< 0.001	1	0.0135	< 0.001
75%	< 0.001	0.0135	1	< 0.001
100%	0.3862	< 0.001	< 0.001	1

Table 2: Pairwise Welch t-test p-values for Table 1 (Drag, 10° AoA)

Trial	Baseline	50% Placement	75% Placement	100% Placement
1	3.5	3.9	3.9	3
2	3	4.1	3.9	3.1
3	3.2	4	3.7	3.3
4	2.9	3.8	3.9	3
5	3.1	4	3.9	3.1
6	3	3.9	4.1	3
7	3.2	3.9	4.2	3.2
8	3	3.6	3.8	3.1
9	3.1	4	4	3.3
10	2.9	3.8	3.9	3
Mean	3.1	3.9	3.9	3.1
SD	0.2	0.1	0.1	0.1

Table 3: Drag Produced (V) from Placement Types at a 0° Angle of Attack



Placement	Baseline	50%	75%	100%
Baseline	1	< 0.001	< 0.001	0.773
50%	< 0.001	1	0.6414	< 0.001
75%	< 0.001	0.6414	1	< 0.001
100%	0.773	< 0.001	< 0.001	1

Table 4: Pairwise Welch t-test p-values for Table 3 (Drag, 0° AoA)

Trial	Baseline	50% Placement	75% Placement	100% Placement
1	5.8	6.1	5.7	5.9
2	6	6.1	5.8	5.8
3	5.7	5.9	5.7	5.9
4	5.9	6.1	6.2	5.9
5	6.1	6.2	6	5.7
6	5.8	6	5.9	5.8
7	5.5	5.8	6.1	5.7
8	5.9	5.7	5.8	5.7
9	5.2	5.8	5.8	5.9
10	6.2	6	5.9	5.8
Mean	5.8	6	5.9	5.8
SD	0.3	0.2	0.2	0.1

Table 5: Drag Produced (V) from Placement Types at a -10° Angle of Attack



Placement	Baseline	50%	75%	100%
Baseline	1	0.153	0.4641	1
50%	0.153	1	0.2926	0.0166
75%	0.4641	0.2926	1	0.2003
100%	1	0.0166	0.2003	1

Table 6: Pairwise Welch t-test p-values for Table 5 (Drag, -10° AoA)

Trial	50% Placement	75% Placement	100% Placement
1	0.12	0.08	0.09
2	0.09	0.07	0.11
3	0.1	0.03	0.05
4	0.07	0.08	0.06
5	0.11	0.08	0.03
6	0.08	0.07	0.07
7	0.11	0.09	0.05
8	0.12	0.07	0.05
9	0.09	0.1	0.05
10	0.07	0.08	0.04
Mean	0.1	0.08	0.06
SD	0.02	0.02	0.02

Table 7: Energy Produced (V) from Placement Types at a 10° Angle of Attack



Placement	50%	75%	100%
50%	1	0.0218	0.0017
75%	0.0218	1	0.1358
100%	0.0017	0.1358	1

Table 8: Pairwise Welch t-test p-values for Table 7 (Energy, 10° AoA)

Trial	50% Placement	75% Placement	100% Placement
1	0.25	0.17	0.09
2	0.2	0.23	0.06
3	0.17	0.18	0.1
4	0.29	0.2	0.09
5	0.27	0.16	0.11
6	0.17	0.13	0.08
7	0.3	0.16	0.06
8	0.24	0.12	0.06
9	0.19	0.15	0.05
10	0.23	0.11	0.09
Mean	0.23	0.16	0.08
SD	0.05	0.04	0.03

Table 9: Energy Produced (V) from Placement Types at a 0° Angle of Attack



Placement	50%	75%	100%
50%	1	0.0018	< 0.001
75%	0.0018	1	< 0.001
100%	< 0.001	< 0.001	1

Table 10: Pairwise Welch t-test p-values for Table 9 (Energy, 0° AoA)

Trial	50% Placement	75% Placement	100% Placement
1	0.27	0.15	0.05
2	0.28	0.24	0.03
3	0.25	0.16	0.07
4	0.26	0.23	0.06
5	0.24	0.18	0.06
6	0.19	0.23	0.03
7	0.23	0.25	0.08
8	0.25	0.12	0.04
9	0.24	0.2	0.03
10	0.26	0.25	0.06
Mean	0.25	0.21	0.05
SD	0.02	0.05	0.02

Table 11: Energy Produced (V) from Placement Types at a -10° Angle of Attack



Placement	50%	75%	100%
50%	1	0.0153	< 0.001
75%	0.0153	1	< 0.001
100%	< 0.001	< 0.001	1

Table 12: Pairwise Welch t-test p-values for Table II (Energy, -10° AoA)

GRAPHS

Figure 5 compares the mean drag produced (in volts) by each placement type across all three angles of attack, with error bars showing the average standard deviations. The overall order from highest to lowest mean drag was 50%, 75%, 100%, and baseline placement (no sensor attached).

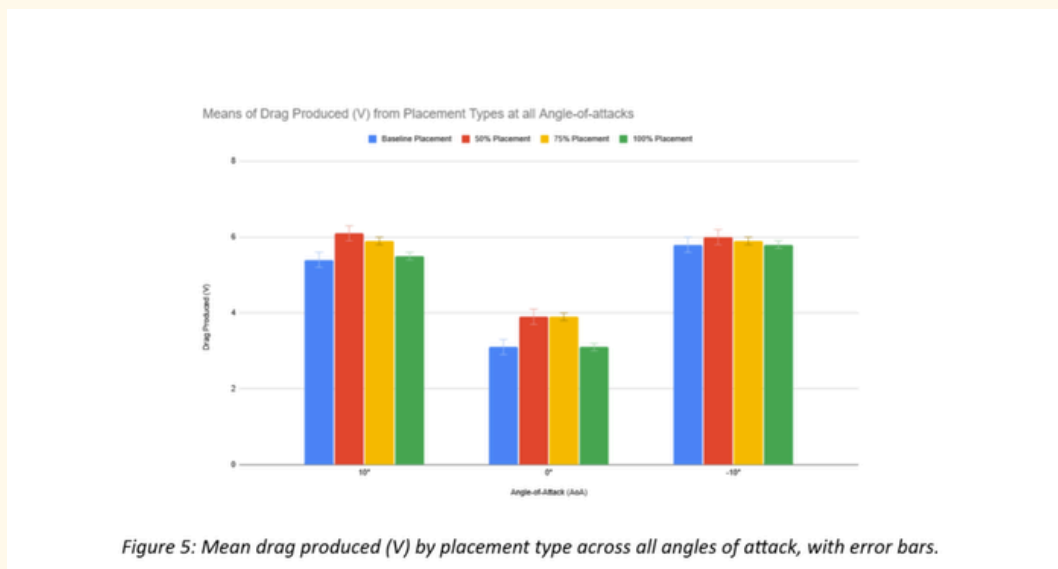
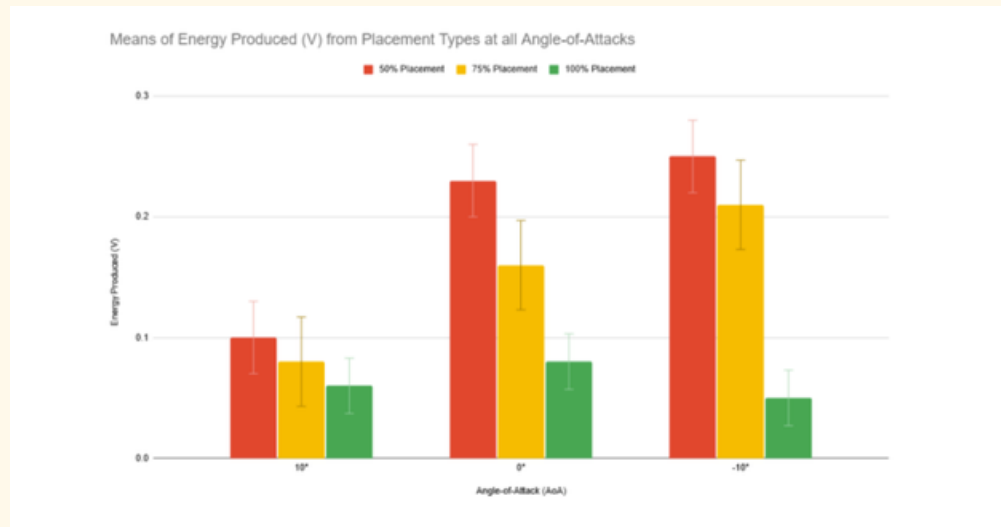


Figure 6 compares the mean energy produced (in volts) by each placement type across all three angles of attack, with error bars showing the average standard deviations. The overall order from highest to lowest mean energy produced was 50%, 75%, and 100%.



STATISTICAL ANALYSIS

Percentage change in drag relative to the baseline was calculated for each placement at each angle of attack by subtracting the baseline mean drag from the placement mean drag, dividing by the baseline mean, and multiplying by 100. Table 13 summarises these results.

Angle of Attack	50% vs. Baseline	75% vs. Baseline	100% vs. Baseline
10°	11.97%	8.47%	1.47%
0°	26.21%	27.18%	0.65%
-10°	2.75%	1.38%	0.00%

Table 13: Percentage Change in Drag Produced Relative to Baseline

These results show that moving the sensor placement to 50% or 75% of span tended to increase drag relative to the baseline at all angles tested, with the largest relative effect at 0° AoA. The 100% placement produced only a small change in drag at 10° and 0° AoA, and no measurable change at -10° AoA, indicating a negligible aerodynamic effect at the location closest to the wingtip.



Across all three angles of attack tested, mean drag was highest for the 50% placement, at 10° AoA (baseline vs. 50% $p < 0.001$, baseline vs. 75% $p < 0.001$, 50% vs. 75% $p = 0.0135$, 50% vs. 100% $p < 0.001$, 75% vs. 100% $p < 0.001$) and at 0° AoA (baseline vs. 50% $p < 0.001$, baseline vs. 75% $p < 0.001$, 50% vs. 100% $p < 0.001$, 75% vs. 100% $p < 0.001$), with the exception of the comparison between 50% and 75% at 0° AoA, which was not statistically significant ($p = 0.6414$). At -10° AoA, most pairwise comparisons were not statistically significant, with the sole exception of 50% vs. 100% ($p = 0.0166$). For energy output, the mean ordering was consistently $50\% > 75\% > 100\%$ across all three angles of attack. The comparison between 50% and 100% was highly significant at every angle ($p < 0.001$), and 50% vs. 75% was significant at 10° AoA ($p = 0.0218$), 0° AoA ($p = 0.0018$), and -10° AoA ($p = 0.0153$). The comparison between 75% and 100% was significant at 0° and -10° AoA ($p < 0.001$) but not at 10° AoA ($p = 0.1358$).

DISCUSSION

The drag results show that sensor placement meaningfully affects aerodynamic behaviour at higher angles of attack. At 10° , the 50% placement produced the largest increase in drag, followed by the 75% placement, while the 100% placement remained close to the baseline. The 100% placement therefore appears to minimise drag penalty by moving the device away from the region of the wing most sensitive to airflow disturbance at higher angles. At 0° , both the 50% and 75% placements increased drag by a similar amount relative to the baseline, with no statistically significant difference between them ($p = 0.6414$), while the 100% placement again remained close to the baseline. This indicates that at near-level flight conditions, drag is sensitive to whether the sensor is placed near the middle of the wing, but not to small differences within that middle region. At -10° , drag differences between placements were small and, with one exception, not statistically significant, suggesting that at negative angles of attack sensor placement has little effect on drag and allows greater flexibility in placement without compromising aerodynamic performance.

In contrast to drag, energy production showed a clear and consistent ranking across all three angles of attack, with the 50% placement producing the most energy, followed by the 75% placement, and the 100% placement producing the least. These differences were statistically significant in nearly all pairwise comparisons, confirming that placing the sensor closer to the centre of the wing allowed it to experience stronger airflow-induced motion, making it more effective for electrical energy generation. Overall, the results reveal a trade-off between minimising drag and maximising energy production. If minimising drag is the priority, the 100% placement is the preferable choice, as it consistently behaved similarly to the baseline across all angles tested. If maximising energy production is the goal, the 50% placement is the clearest option, though it increases drag at positive angles of attack. The 75% placement offers a compromise, producing significantly more energy than the 100% placement while causing less drag than the 50% placement.



Several sources of error should be considered when interpreting these results. Angle of attack was set and verified manually using a classroom protractor before each trial, which could introduce both systematic and random error in the true AoA tested; because piezoelectric output is sensitive to small changes in bending and flow separation, such error could mask real differences between sensor positions or create a false impression of which configuration produced the best output. Sensor mounting also introduced variability, as double-sided tape allowed for slight shifts in position and pressure between runs, which could alter how the winglet flexed at the sensor location and affect both drag and voltage readings independently of the intended placement variable. The wind tunnel itself introduced further limitations: the requirement to satisfy the 10% blockage rule meant the wing model had to be scaled down substantially, which can cause airflow to separate earlier and behave differently than it would over a full-scale wing, and the tunnel accommodated only one sensor at a time rather than an array. Finally, instrumentation error is a relevant consideration, as piezoelectric disks produce high-impedance, often transient signals that require a charge amplifier or high-impedance data acquisition system for accurate capture; a basic multimeter without amplification may underestimate peak voltages or miss brief voltage spikes, meaning the electrical output recorded here may understate the sensor's true generating capacity.

These findings are useful for guiding further research into energy harvesting and aerodynamic efficiency in aircraft design. This study demonstrated that piezoelectric devices integrated into winglets can generate measurable electrical output under airflow while producing only a minimal increase in drag at certain placements, showing that even small structural deformations can be exploited for low-power energy generation. The results clarify which angles of attack and sensor locations are most promising, allowing future work to focus on the most favourable configurations rather than broad trial-and-error testing. Future work should test different piezoelectric materials, sensor geometries, and array configurations to determine how power output scales with design choices, expand the aerodynamic test scope to quantify the energy cost associated with any increase in drag, integrate power-conditioning electronics, embed sensors into more realistic wing structures, and conduct longer-duration tests to assess durability under repeated cyclic loading.



CONCLUSION

This study investigated whether piezoelectric ceramic sensors integrated onto a winglet could generate measurable electrical output while remaining within acceptable aerodynamic drag limits, testing three sensor placements (50%, 75%, and 100% of wing span) across three angles of attack (10° , 0° , and -10°). The results partially supported the initial hypothesis: the 50% placement did maximise energy output at all angles tested, but it did not achieve this while minimising drag change, particularly at 10° and 0° AoA, where it produced the largest drag penalty of the three placements. The 100% placement consistently behaved most like the baseline wing in terms of drag, making it the preferable choice where minimising aerodynamic penalty is the priority, while the 75% placement offered a balance between the two extremes. These conclusions are subject to the limitations discussed above, including manual AoA verification, tape-mounted sensor variability, wind-tunnel scaling effects, and the use of an unamplified multimeter for voltage measurement, any of which could have affected the absolute values recorded even where the relative trends between placements remained consistent. Overall, this proof-of-concept study provides experimental evidence that piezoelectric energy harvesting on aerodynamic surfaces is feasible and worth further investigation as a supplementary power source for onboard sensors and low-power avionics.





EFFECT OF CARBARYL CARBAMATE ON INSULIN SIGNALING IN 3T3-L1 MOUSE ADIPOCYTE CELLS

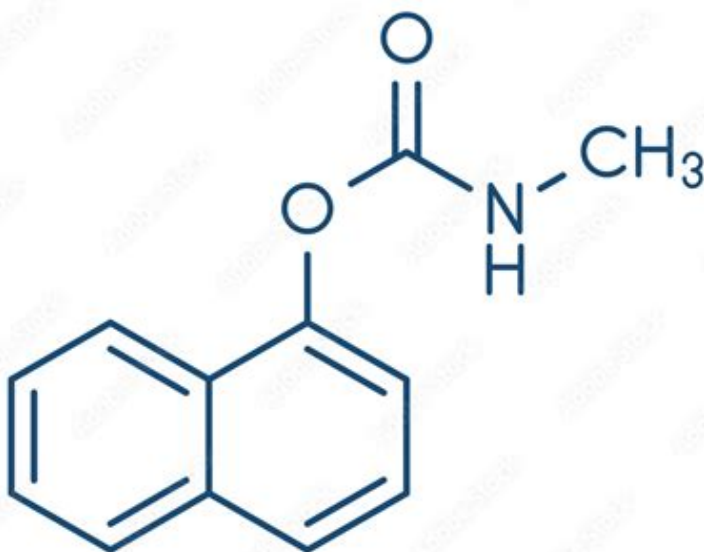
BY UNNATI MISHRA





ABSTRACT

In the U.S., approximately 4 million pounds of carbaryl-based pesticides are used annually in agriculture, with an additional million pounds used in residential settings. Despite widespread exposure, the effects of carbaryl on metabolic health remain poorly understood. While other pesticides have been studied extensively, limited research examines if carbaryl interferes with insulin signaling and potentially contributes to conditions like Type II diabetes. Previous studies suggest carbaryl may disrupt insulin receptors, but lack definitive conclusions. This study investigated whether carbaryl exposure alters the insulin signaling pathway in NIH-3T3 L1 cells. The final step of this pathway involves phosphorylation of the protein Akt. It was hypothesized that carbaryl exposure would significantly reduce phosphorylated Akt (p-Akt) levels compared to untreated cells. Fibroblast cells were seeded in 6-well plates and differentiated into adipocytes to express insulin receptors. Two control groups and two experimental groups were established. One control received no treatment, while the second received 100 μ M insulin for 20 minutes. Experimental groups were exposed to either 150 nM or 350 nM carbaryl for 3 hours prior to insulin stimulation. Samples were collected using trypsin, snap-frozen, and analyzed using Western blotting for Akt and p-Akt with tubulin normalization. The p-Akt blot failed to detect phosphorylation. However, normalized results showed reduced total Akt levels in carbaryl-treated samples compared to the baseline control. These findings suggest carbaryl may affect earlier stages of signaling by altering protein expression. Future studies should use larger culture platforms to increase cell yield and protein recovery for more reliable quantification.



carbaryl



INTRODUCTION

Pesticide Usage, Carbamates, & Carbaryl

The United States Environmental Protection Agency defines pesticides as "Any substance or mixture of substances intended for preventing, destroying, repelling, or mitigating any pest." (U.S. Environmental Protection Agency, n.d.). The usage of pesticides and insecticides more than doubled in the last half of the 20th century, and has continued to rise in prevalence in the 21st century as well (Figure 1).

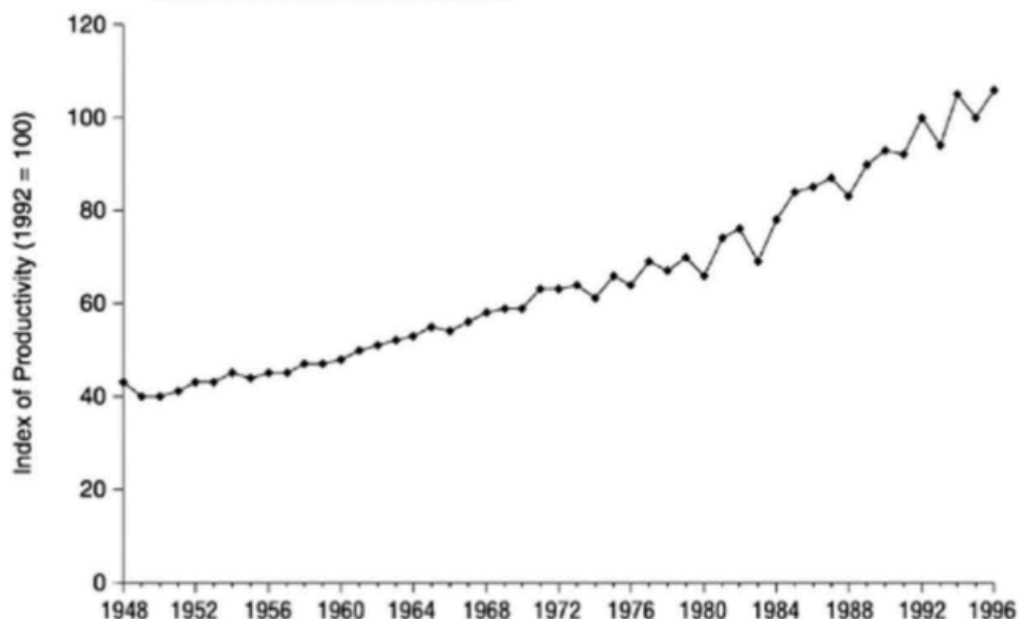


Figure 1. The increase in pesticide usage across 1948-1996. Usage more than doubled in this time frame, and has continued to grow in the 21st century as well. (National Academies, 2000).

Pesticides are vital to modern trade and food supply; by making agricultural practice more efficient, countries are better suited to address rising food demand corresponding with continually rising populations and also participate in world trade (Aktar et al., 2009). They hold an undeniable role in the global economy and addressing urgent world issues such as hunger and poverty. However, their widespread usage consequently demands extensive research to ensure they are safe for exposure to non-target organisms.

Carbamates are a category of organic compounds including Aldicarb, Bendiocard, Carbaryl, Dioxacarb, Fenobucarb, Isoprocarb, and Methomyl that are commonly used in pesticides (Mdeni et al., 2022). Their most common use is in agriculture, accounting for 6% of global insecticide usage. Because carbamates are rapidly metabolized by mammals, they are often assumed to exert only moderate biological effects, leading to their widespread usage without equivalent toxicological scrutiny (Figure 2).

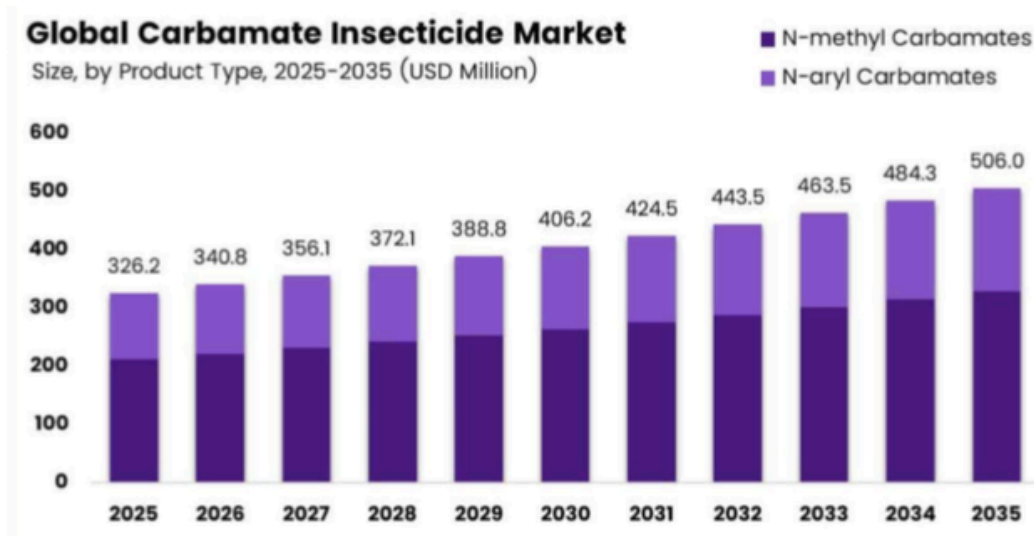


Figure 2. Carbamate usage is projected to continue increasing in the next decade. Carbaryl is an N-methyl carbamate, depicted in dark purple. (market.us, 2026)

Carbaryl is a chemical commonly found in pesticides, often under products labeled as Sevin, ranging from Sevin dust to liquid products. Due to its widespread availability in most gardening stores, it has become commonly used in residential and commercial areas to eliminate ants, aphids, caterpillars, fleas, spiders, and more for agricultural or aesthetic purposes. The chemical functions as an inhibitor for acetylcholinesterase, an enzyme that breaks down accumulation of the neurotransmitter acetylcholine. As the degradation of acetylcholine in the synaptic cleft is decreased, its accumulation causes neurotoxicity. Thus, when exposed to plants, it effectively kills pests by inducing neural damage, nerve overstimulation, and muscle spasms (Bond et al., 2016).

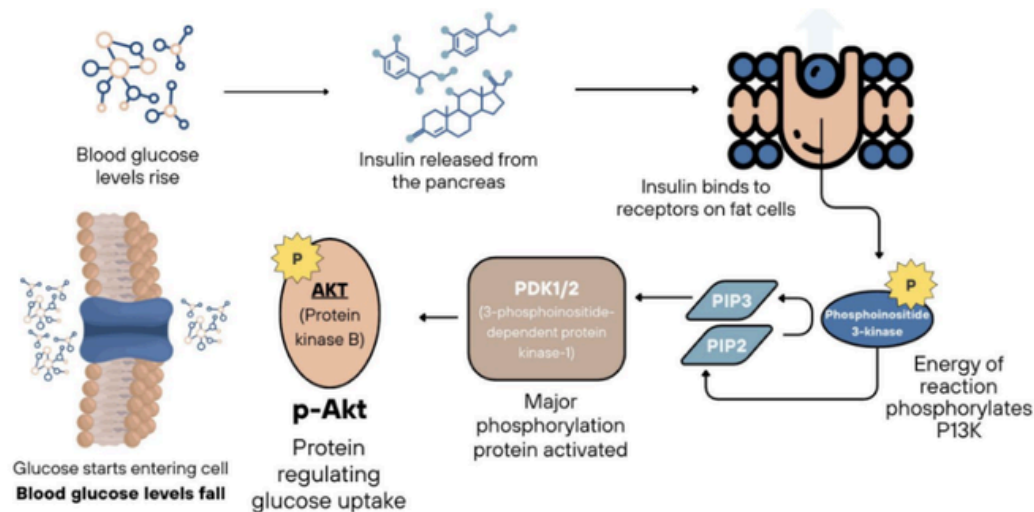
Research has noted the ability of carbaryl to affect larger mammals including humans in the same way. An early study by Best & Murray (1962) analyzed the effects of various times of exposure to Sevin dust, finding that it is metabolized rapidly in the body determined by frequent blood cholinesterase activities. Those more heavily exposed to Sevin had more adverse effects, with cholinesterase inhibition being the most severe symptom. Thus, it is evident that the health impacts of carbaryl have been explored for decades, but given its common usage today, not enough findings have conclusively demonstrated the extent to which it could pose a potential danger to the long term health of non-target organisms.

Certain administrations have noted the dangerous potential of this chemical and acted to avoid its usage; in 2020, the California Department of Pesticide Regulation ruled to restrict general consumer use of carbaryl-based pesticides (Cassidy, 2020), noting that general residents do not have the training to properly protect themselves and non-target organisms upon using the pesticide. Carbaryl products have since been labeled as "restricted materials," only used by licensed individuals working in agricultural settings. California's inhibition of carbaryl usage demonstrates that the chemical poses a significant risk to human and environmental health; upon further detailed research about carbaryl's health effects, more locations may follow suit. But without a proper understanding of the extent to which carbaryl can affect mammal bodily functions, public health standards cannot be effectively met.



Insulin Signaling Pathway

The insulin signaling pathway is one of the most vital bodily functions, maintaining glucose homeostasis in the body (Picture 1). As food enters the small intestine, it is broken into glucose released into the bloodstream. This stimulates the pancreas to release the hormone insulin into the bloodstream to clear glucose from the blood and store it as glycogen, lipids, or ATP. In fat cells, referred to as adipocytes throughout this paper, insulin binds to specific insulin receptors, triggering protein phosphorylation. After a sequence of phosphorylation cascades, one of the last proteins to be phosphorylated is the enzyme Akt, which then regulates glucose uptake and metabolism by phosphorylating downstream targets.



Picture 1. A simplified demonstration of the insulin signaling pathway. After glucose levels rise, insulin is secreted from the pancreas and binds to insulin receptors on fat cells. The energy of this reaction releases energy in the form of a phosphate group, triggering a “phosphorylation cascade” where various proteins are phosphorylated, or activated, to perform their intended function.

Insulin resistance occurs when downstream components of the insulin-signaling pathway inhibit upstream elements, or when unrelated pathways inhibit the signaling pathway. Similarly, if another chemical blocks insulin's interaction with insulin receptors in adipocytes, the signaling pathway is not triggered, and proteins are not phosphorylated. When this occurs, ordinary levels of insulin are not able to interact with the receptor to start glucose uptake and storage, leading to conditions such as Type 2 Diabetes where patients are in need of insulin injections to reduce the amount of glucose in their bloodstream.

Phosphorylated-Akt, referred to as p-Akt throughout this paper, serves as an effective measurement of successful phosphorylation throughout the insulin signaling pathway, as it is the final protein directly activated in the pathway. If Akt is not phosphorylated, Glu4 channels in the membrane that allow glucose to enter the cell do not form, and blood glucose levels remain high. This study aimed to identify the amount of p-Akt and Akt expressed shortly after insulin stimulation after various treatments of carbaryl.

Notably, insulin resistance can occur even if Akt is phosphorylated; downstream factors may not properly be activated even if p-Akt is present. Thus, the results of this study only suggest whether or not carbaryl impacts the phosphorylation of Akt and the presence of Akt itself, rather than conclusively determining if insulin signaling is disrupted.



Insulin Pathways & Pesticides

Other chemical groups commonly used in pesticides have been observed to interfere with insulin signaling pathways in mammal cells. As explored by Xiao et al. (2017), there is a growing body of evidence showing that insecticide exposure can elevate risk of obesity and Type 2 diabetes by interfering with insulin signaling pathways. Extensive studies on chemicals including organochlorine and organophosphorus linked them to mammalian insulin-related conditions, but as Xiao et al. note, there is a paucity of studies relating pesticides that contain carbamates including carbaryl and their potential impacts on insulin signaling.

Aside from cholinesterase inhibition upon exposure to various doses of carbaryl, other negative health effects have been documented as well, including speculation regarding description of insulin secretion. Sahana & Agarwal (2018) report that carbaryl can damage normal liver functions; by disrupting acetylcholinesterase, the buildup of acetylcholine can potentially impact insulin secretion from the pancreas. This builds upon the findings of Best & Murray, suggesting that acetylcholinesterase disruption affects other bodily functions and systems as well. Notably, both studies suggest more of a correlation between carbaryl and negative health impacts, emphasizing the need for more purposeful research directly examining the effects of carbaryl on the body.

Carbaryl has previously been suggested to impact insulin pathways, but no direct experiments have strengthened the potential correlation. A 2014 study by Popovska-Gorevski et al. analyzed whether carbaryl can bind to MT2 melatonin receptors due to their structural similarity with insulin receptors. Although this study used Chinese hamster ovary cells rather than pancreatic or adipocyte models, its implications for metabolic research are important – the authors demonstrate that carbaryl is structurally similar enough to melatonin to dock with the MT2 receptor model. MT2 receptors are known to be structurally similar to insulin receptors in mammals, so this docking suggests a plausible mechanism by which carbaryl could interfere with insulin release in mammalian systems. Yet notably, due to the chemical delivery to melatonin receptors rather than insulin, the results of this study cannot draw solid conclusions about carbaryl's influence on actual insulin receptors in mammals. Thus, there is no conclusive evidence to suggest that carbaryl interferes with mammalian insulin receptors by affecting the insulin signaling pathway.

3T3-L1 Cells

3T3-L1 is a commonly used line of embryonic mouse cells. They are initially in the form of fibroblast cells, a type of connective skin tissue, but 3T3-L1 cells are commonly used in insulin signaling studies due to their ability to differentiate (see Appendix A) into adipocytes that express insulin receptors (American Type Culture Collection, n.d.); thus, this line is well suited to studying cellular mechanisms for diabetes, obesity, and other relations of the insulin signaling pathway.



Gap Justification & Relevance

Residential usage of pesticides is often overlooked due to its commonality. However, research by Hosni et al. (2024) discovered that signal words commonly found on pesticide and insecticide warning labels fail to adequately communicate toxicity levels. Analyzing incentivized eye tracking on conventional warning labels, they suggest that the current labeling system may lack informative power and have unintended harmful effects on human health when used by unknowing residents in areas exposed to other humans and non-target organisms including pets and animals.

It is more important than ever to identify the human health hazards associated with pesticide usage and limit its prevalence in residential areas for the health of those in proximity to exposure. In particular, to avoid unintentional development or heightening of insulin-related conditions including Type 2 diabetes, the market for which is projected to almost double in the next decade (Biospace, 2024), identifying potential causes or contributors to insulin disruption is vital to prevention of these conditions. He et al. (2020) additionally emphasize that insulin resistance is a common side effect of eating food made with pesticides that, unlike organophosphorus and organochlorine, lack research about its impacts on the signaling pathway.

No studies have evaluated carbaryl exposure in NIH3T3-L1 adipocytes for the purpose of analyzing insulin reactivity. The gap in the current literature indicates a strong need for research that directly assesses the impact of carbaryl on the insulin signaling pathway in mammalian cells, particularly adipocytic fat cells that effectively model glucose homeostasis. This research aims to address this gap and contribute to the body of research about the health consequences of carbaryl.

Research Question and Hypothesis

The purpose of this research is to identify if there is a significant difference between the amount of p-Akt and Akt after insulin stimulation when cells are administered a carbaryl concentration. This research aims to answer the question: does exposure to carbaryl impact the insulin signaling pathway leading to insulin resistance in 3T3-L1 cells? Given that carbaryl has been suggested to bind to receptors structurally similar to insulin receptors in 3D modeled research, I hypothesize the carbaryl will significantly decrease the levels of phosphorylated-Akt and Akt compared to cells not treated with carbaryl.

METHODOLOGY

Overview

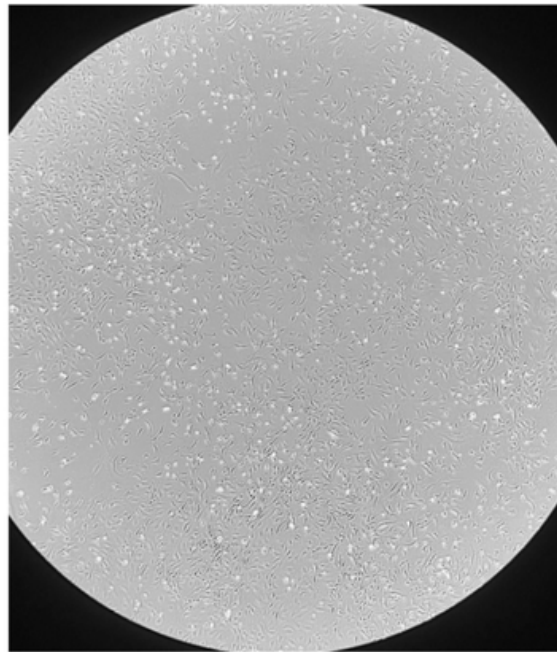
T3T3-L1 cells were initially undifferentiated fibroblast mouse cells that were cultured in T25 adhesion flasks (see Appendix A), expanded in T75 adhesion flasks (see Appendix A) then seeded into 6-well plates (see Appendix A) to undergo a differentiation process to ensure expression of insulin receptors as adipocytes (Picture 2). Once differentiation was visually confirmed, at least 3 wells were treated with a 150 uM carbaryl treatment or a 350 uM carbaryl treatment before insulin stimulation, while at least 3 others were treated with only insulin and others received no chemical interference. Samples were accumulated into pellets (see Appendix A) and a snap freeze (see Appendix A) was performed to preserve cellular activity before a Western blot assay was performed to quantify levels of Akt and p-Akt in each group. Data was analyzed using ImageJ software (see Appendix A).



Picture 2. Some of the T75s and 6-well plates I seeded. The T75s were intended to continue growing a healthy amount of cells, while the 6-well plates were the first effort at differentiating cells.

Cell Culture

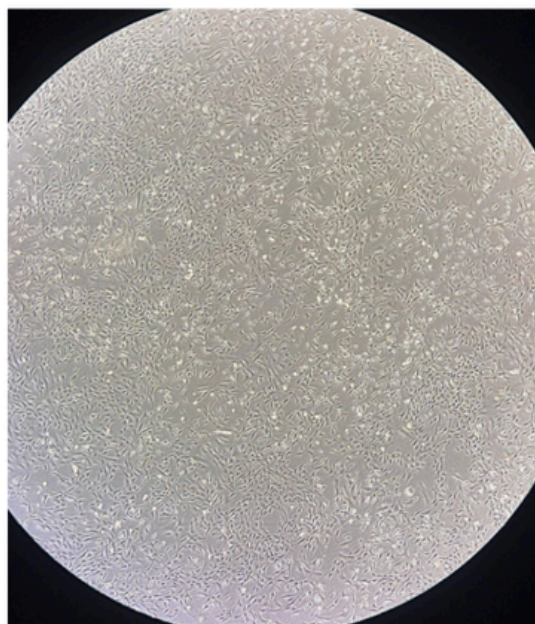
Five vials of 3T3-L1 cells, donated from University of Colorado Anschutz Medical Campus, were kept in cryofreeze inside a tank of liquid nitrogen. For thawing procedures, protocols from Editgene (n.d.) were utilized despite being labeled for regular NIH3T3 cells, because the L1 cell line is cultured very similar to the parent line before differentiation. First, each cryovial was thawed and initially cultured in a T25 flask using Basal Media (BMI) (see Appendix A), of High Glucose Dulbecco's Modified Eagle Medium, newborn calf serum, and penicillin-streptomycin. Newborn calf serum provided essential nutrients and growth factors and the antibiotic was necessary to prevent potential bacterial infestation in the culture. Once ~80% confluence (see Appendix A) (Picture 3) was visually confirmed, cells were passaged (see Appendix A) into T75s to continue growing the amount of cells I had available before trials began.



Picture 3. A T25 at ~80% confluence underneath the microscope. The plate is still visible, although largely covered by cells.

Pre-Trials

While my initial goal during pre-trials was to successfully differentiate cells, upon thawing the first cryovial, my primary issue became determining proper cell culture methods without overgrowth. The first T25, labeled 1A, reached confluency in roughly 4 days, but when it was passaged into T75 plate 1B, the calculated proportions to seed 0.3×10^6 cells into each well (Thermofisher, n.d.) resulted in too high of a concentration; within one day, each well appeared around 95% confluence (Picture 4). I concluded that the calculated seeding concentration was far too high, and shifted my focus from differentiation to properly seeding cells.



Picture 4. A T25 at ~95% confluence underneath the microscope. Most of the plate is covered by cells. This amount of growth so fast was unexpected, and I had to adjust my timeline to work faster and passage cells more often.

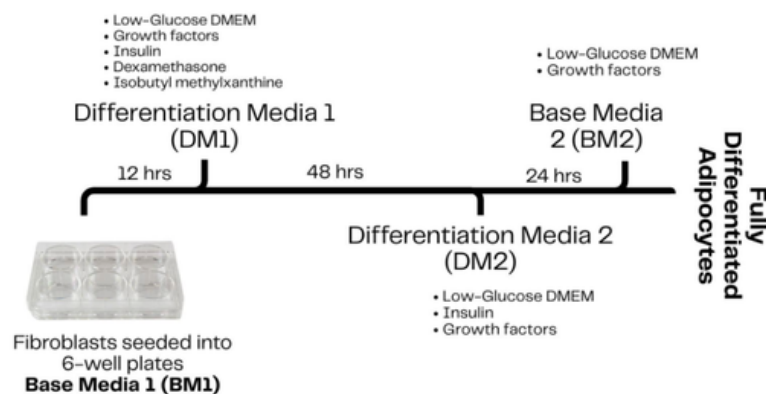


When passaging plate 1B, I was unsure if cells would remain viable (see Appendix A) since 1B had reached a much higher confluency than recommended. To determine the best seeding density for prolonged growth, T75 1C was seeded with 0.35 mL cell concentrate, plate 1D was seeded with 0.1 mL cell concentrate, and plate 1E was seeded with 0.175 mL cell concentrate.

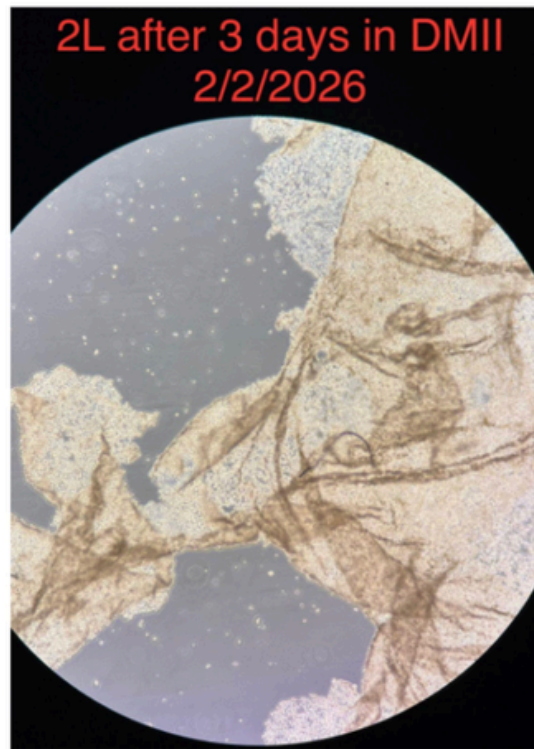
After two days of growth, 1C reached roughly 85% confluency, growing from 55% to 85% confluency in two days. Many cells appeared stressed, looking compacted; I concluded that the seeding cell concentration of 0.35 mL was likely too high. Plate 1D appeared to be around 60% confluency, growing from ~25% to 60% confluency in two days. The seeding cell concentration of 0.1 mL was deemed suitable if wanting to culture cells for an extended period of time. Finally, Plate 1E seemed to be around 75-80% confluency, growing from ~40% to 75% confluency in two days; the seeding cell concentration of 0.175 mL was good for seeking to culture cells to confluency within 2-3 days.

Differentiation Into Adipocytes

Following protocols from Zhao et al. (2019), three new differentiation media were created. Base Media II (BMII) (see Appendix A) used Low-Glucose DMEM, FetalGro (see Appendix A), and penicillin-streptomycin. Differentiation Media I (DMI) (see Appendix A) was made by adding differentiation factors and bovine insulin to BMII. Finally, Differentiation Media II (DMII) was made by only adding insulin to BMII. Picture 5 demonstrates the differentiation process as suggested by Zhao et al. (2019) adjusted for my findings; while Zhao et al. suggested longer exposure times, I found exposure to DMII for too long resulted in overgrowth and cell death (Picture 6) and adjusted the timeline accordingly.



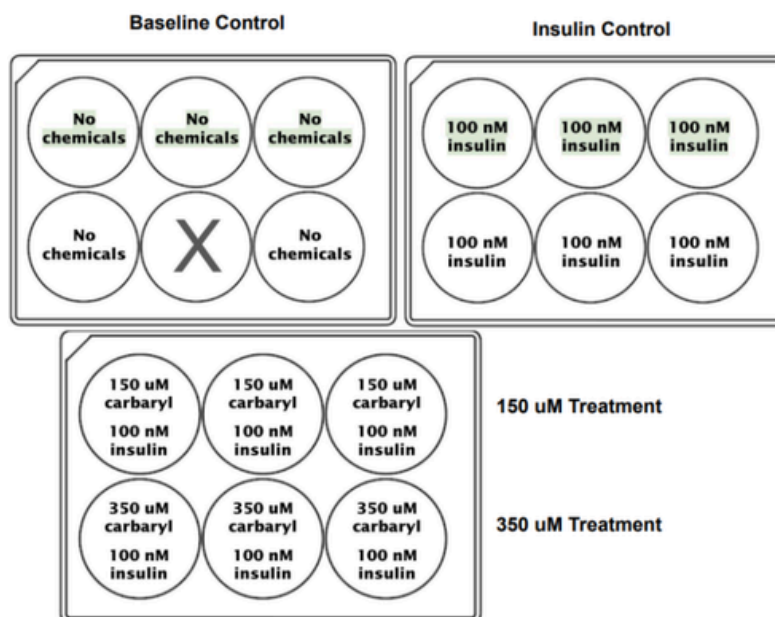
Picture 5. A timeline depicting the process of differentiation. While the timeline for each media suggested by Zhao et al. (2019) was longer, because my cells multiplied so quickly, this is the time frame I determined to be most effective in differentiation. See Appendix A for more information on exact concentrations of chemicals in each media.



Picture 6. Protocols from Zhao et al. (2019) suggested an exposure time of 2-3 days in DMII, but I found that cells repeatedly overgrew and died in this

Sample Groups

Four sample groups were created (Picture 7). Three biological replicates per group are commonly used as a minimum in cell-culture Western blot studies, so my goal was to collect at least three samples from each group.



Picture 7. Each experimental and control group. Three wells of the Baseline Control and Insulin Control respectively are highlighted in green; this is because I had collected samples for the entire 6-well plate, but could only assay 3 samples for each group due to the limited Western blots available, so the green depicts which wells were used in the Western blot.



Before administering any chemicals or collecting samples, all groups were exposed to a Starvation Media (see Appendix A) for 24 hours in order to minimize any residual chemical effects from the contents of the media the cells were kept in. This also heightened the cells' response to the chemicals that would be administered to them. Samples for the Baseline Control were collected immediately after these 24 hours, while all other groups continued to receive various chemical treatments.

Carbaryl Administration

A 2017 study by Ferruccio et al. analyzed the effects of carbaryl, sun exposure, or both on human skin cells, testing many different carbaryl concentrations. They found that 339 μM of carbaryl in DMSO (see Appendix A) is lethal to cells, and used 100 μM for experimentation, administering the chemical through the cells' media. However, in 2016, Xu et al. exposed PC12 rat cells to carbaryl attempting to assess biochemical changes, using a concentration of 500 μM in DMSO. Similar to Ferruccio et al., carbaryl was added to the cells' media in order to administer it to them. Because these concentrations differ, I used these protocols as an upper and lower bound reference for carbaryl concentrations. Three wells of a 6-well plate were treated with 150 μM carbaryl in DMSO (150 μM Treatment), while the other three wells were treated with 350 μM carbaryl in DMSO (350 μM Treatment), allowing a representation of lower, more moderate concentrations of carbaryl compared to more drastic concentrations.

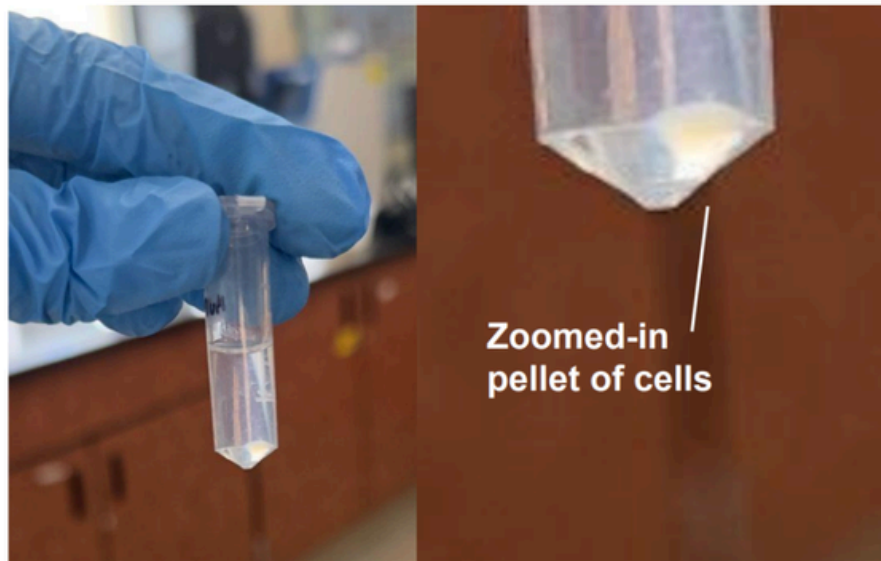
Cells were exposed to carbaryl and kept in the incubator for 3 hours as suggested by the aforementioned studies.

Insulin Stimulation

As suggested by Furukawa et al. (2019) at Harvard Medical School, an insulin concentration of 100 nM in DMSO was administered to the cells for roughly 20 minutes. The Insulin Control group received this stimulation immediately after 24 hours in Starvation Media, while the 150 μM Treatment and 350 μM Treatment groups received stimulation after 3 hours of their respective carbaryl exposures.

Sample Collection

Cell media was aspirated (see Appendix A) from each well of each group and replaced with trypsin (see Appendix A), which was then neutralized using Starvation Media. The contents of each well were pipetted into a centrifuge tube labeled with the group name, and centrifuged at 10,000 rpm for 5 minutes to form a cell pellet (Picture 8).

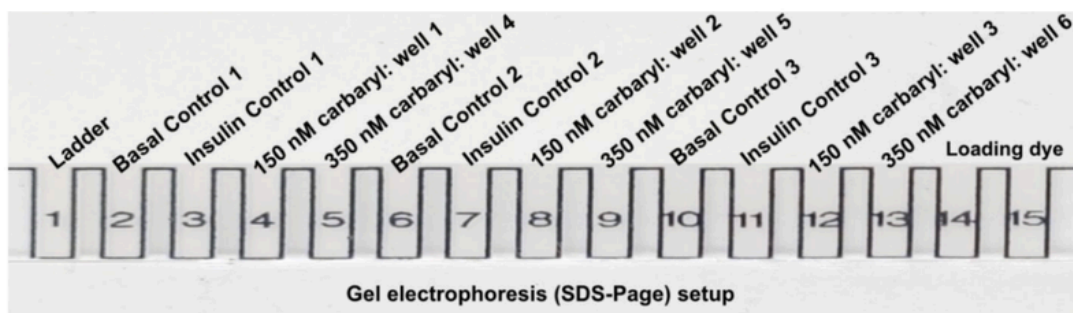


Picture 8. Each cell pellet consisted of the cells in one well of a 6-well plate. In many studies, the cell pellet is far larger due to using different seeding platforms (such as petri dishes), but due to the small scale of my high-school level project, I had to collect samples in 6-well plates, resulting in the small size of the cell pellets.

The supernatant (see Appendix A) was aspirated and the cell pellet was then resuspended (see Appendix A) in a retention buffer (see Appendix A) before the same centrifuge cycle. The final supernatant was aspirated, and cells were preserved using a snap freeze protocol.

Western Blot

The purpose of a Western blot is to quantify amounts of a certain protein in a cell; this assay was performed at the University of Colorado Anschutz Medical Lab. First, each thawed cell pellet was resuspended in a lysis buffer (see Appendix A) and centrifuged at 21,000 g for 10 minutes to separate DNA from the proteins. The supernatant contained proteins, and this was transferred to a newly labeled spin-filter tube (see Appendix A) while the remaining DNA pellet was disposed of as biohazard waste. In the spin-filter tubes, samples were centrifuged at 7,500 g for 5 minutes to fully isolate proteins before loading dye (see Appendix A) was added. Half of the contents of each sample were loaded into one gel for gel electrophoresis (see Appendix A) intended to quantify p-Akt, while the other half was loaded into another gel to quantify Akt (Picture 9).



Picture 9. There are 15 wells in a gel for gel electrophoresis. The first is a ladder, which contains a variety of proteins of known kilobasepairs so intended proteins can be identified depending on where they end up on the gel. Three samples of each group were loaded into 12 wells. The remaining 2 wells were filled with loading dye simply because the lack of pressure from two empty wells might cause the proteins to travel in a skewed manner rather than straight down the gel.



After gel electrophoresis, gels were transferred onto membranes and kept in a transfer buffer (see Appendix A) before one membrane was placed in a solution of buffer and Akt primary antibodies and the other was placed in a solution of buffer and p-Akt primary antibodies (see Appendix A). The next day, membranes were washed off the primary antibodies using a buffer, and then exposed to their respective secondary antibodies (see Appendix A) to visualize the proteins in the membrane. After 45 minutes, blots were imaged and analyzed using ImageJ software.

To account for differences in the amounts of cells in each sample, normalization (see Appendix A) is required. Tubulin is a commonly used protein present in almost all eukaryotic cells, and because it's usually present at relatively constant amounts across samples, it is commonly used as a control in Western blots.

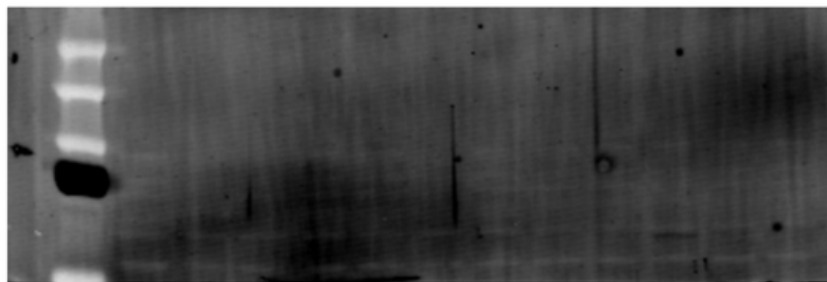
Data Visualization

Using ImageJ, I quantified the amounts of Akt/p-Akt in each sample, and then quantified the amounts of tubulin in each sample, and divided the amounts of Akt/p-Akt by tubulin to determine the amount of the target protein present in each sample.

RESULTS

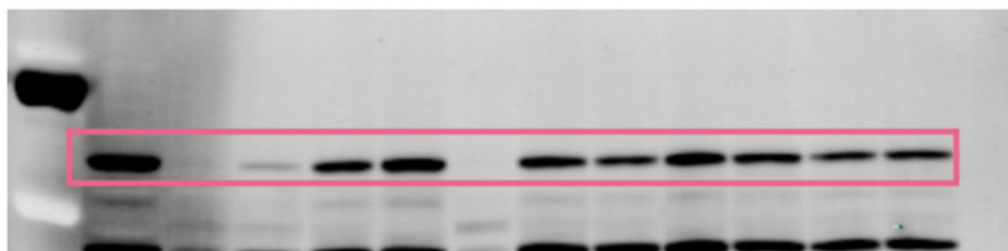
In this experiment, 150 μ M or 350 μ M of carbaryl was administered to adipocytic cells for several hours before insulin stimulation to identify differences in the insulin signaling pathway due to the chemical.

The blot intended to detect levels of phosphorylated-Akt did not result in sufficient amounts of protein to be visualized (Picture 10). Data and conclusions cannot be drawn using this blot.



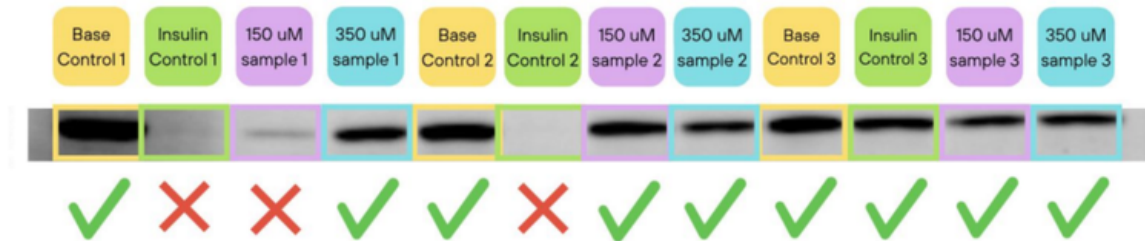
Picture 10. Bands for p-Akt were not sufficient to be visualized.

The blot detecting levels of regular Akt resulted in protein bands that could be mostly visualized (Picture 11). More prominent bands qualitatively correlate with higher levels of protein detection.



Picture 11. The location of Akt bands in the protein display of the Western blot. Notably, in Western blots, it is typical that only the proteins intended should visualize into bands; the presence of other bands lower in the image suggests protein degradation. This occurs when, during the cell lysis step of Western blot preparation, protein enzymes start breaking down all proteins in the cell before the enzymes are removed.

However, data from Insulin Control 1, Insulin Control 2, and 150 uM Treatment 1 did not result in sufficient protein amounts for quantification (Picture 12). As previously stated, Western blot analysis typically utilizes at least three replicates of visible data for a group to accumulate sufficient evidence to draw conclusions about that group. Data from the 350 uM Treatment and Baseline Control was instead quantified and compared.

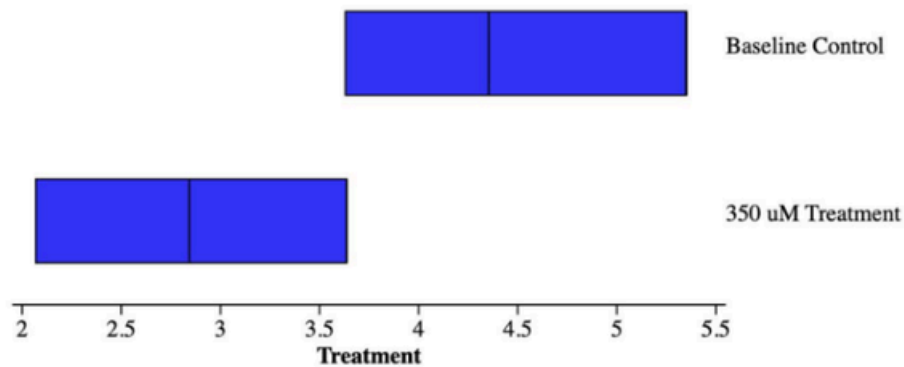


Picture 12. Bands for Insulin Control 1, 2, and 150 uM Treatment 1 were unable to be visualized.

Upon normalization to tubulin, the Baseline Control samples resulted in an average of 4.4453 Akt while the 350 uM Treatment samples resulted in an average of 2.8496 Akt (Graph 1). A two-sample t-test for a difference in means (Graph 2) was performed to identify if the difference between these values was statistically significant. A significance level of 0.05 was utilized, and a p-value less than this number can be considered statistically significant. The t-test resulted in a p-value of 0.038712, less than 0.05, and was thus significant.

Group	Tubulin	Akt	Akt/Tubulin	Average	St. Deviation
Base Control	2054.426	7461.468	3.631899129	4.445340101	0.8629616921
Base Control 2	1674.083	8957.175	5.350496361		
Base Control 3	1872.083	8150.347	4.353624813		
Insulin Control 1	x	x	x	1.457906703	Not enough data
Insulin Control 2	x	x	x		
Insulin Control 3	4954.518	7223.225	1.457906703		
150 uM Treatment 1	x	x	x	31.00863755	15.09040836
150 uM Treatment 2	188.092	7839.518	41.67916764		
150 uM Treatment 3	260.92	5306.619	20.33810747		
350 uM Treatment 1	1846.669	6717.225	3.637481866	2.849616832	0.7845847493
350 uM Treatment 2	2095.991	5958.933	2.843014593		
350 uM Treatment 3	2896.698	5991.397	2.068354036		

Graph 1. The data collected from the Insulin Control and 150 uM Treatment groups cannot be used in data analysis because three replicates of data from each group is commonly known as a minimum amount of data necessary to draw conclusions about a group from a Western blot. The Western blot only successfully quantified one Insulin Control sample and two 150 uM Treatments.



Summary Statistics

Group Name	n	mean	SD	min	Q ₁	med	Q ₃	max
1: Baseline Control	3	4.445	0.863	3.632	3.632	4.354	5.35	5.35
2: 350 uM Treatment	3	2.85	0.785	2.068	2.068	2.843	3.637	3.637

Graph 2. A visual representation of the differences between the means of the Baseline Control and 350 uM Treatments.

DISCUSSION

This research aimed to determine whether various concentrations of carbaryl impact the presence of Akt and p-Akt, essential proteins in the insulin signaling pathway of adipocytic cells. I hypothesized that the levels of Akt and p-Akt would be decreased after exposure to carbaryl compared to groups exposed to only insulin or no chemicals. Fibroblast cells were seeded into 6-well plates and differentiated into adipocytes, and three samples of each group were collected. A total of 6 wells were exposed to either 150 uM or 350 uM carbaryl for 3 hours prior to 20 minutes of insulin stimulation, while 3 wells received only 20 minutes of insulin stimulation and the other 3 received no chemicals. Samples were preserved through snap freeze and proteins were quantified using a Western blot.

Analysis of Results

Due to the lack of sufficient protein visualization in the p-Akt blot, which was necessary to demonstrate whether carbaryl directly blocks insulin reception as suggested by Popovska-Gorevski et al. (2014), conclusions cannot be drawn using that data. My results cannot indicate whether carbaryl binds to the insulin receptors on adipocytic cells.

All conclusions were thus drawn using the 350 uM Treatment and Baseline Control data from the Akt blot. These results could instead indicate whether or not the presence of carbaryl prior to insulin stimulation affects the efficiency of the insulin signaling pathway.



The statistically significant decrease in Akt amounts in cells treated with carbaryl before insulin stimulation compared to baseline amounts present in cells suggests that carbaryl exposure can reduce the abundance of proteins essential to the insulin signaling pathway. The Baseline Control indicates how much Akt is present in adipocytes in a healthy state, without chemical interference or even activation of the insulin signaling pathway. The 4.4453 quantification of Akt thus suggests that the cells comparatively have more Akt present, so if they were to experience stimulation by insulin binding, Akt can regulate the pathway and allow it to carry out its intended function, triggering further downstream reactions that allow glucose to enter the cell. Comparatively, when treated with 350 uM carbaryl, cells might experience chemical stress (see Appendix A) that causes more cellular energy to be spent on survival functions and less to be spent on producing proteins for the insulin signaling pathway. Thus, after insulin stimulation, amounts of Akt were significantly lowered, resulting in 2.8496 Akt, which would make these carbaryl-treated cells far less efficient in completing the signaling pathway.

Limitations & Sources of Potential Error

Without quantification of Akt in the Insulin Control samples, there is not sufficient evidence to compare Akt in the insulin pathway after carbaryl stimulation to amounts of Akt in insulin-stimulated cells without external chemical interference. Further, without Akt quantification in the 150 uM Treatment samples, I cannot draw comparisons regarding Akt amounts after exposure to a smaller concentration of carbaryl.

Had I obtained data regarding the presence of Akt in adipocytes after a regular interval of insulin stimulation (Insulin Control), I would be able to compare how much Akt should be present in these cells after a regular insulin stimulation to how much Akt was present after a cycle of the insulin signaling pathway post exposure to carbaryl. This would allow for a more direct comparison of carbaryl's impact on insulin-stimulated adipocytic cells, because this current data is only able to compare its impact to cells that did not undergo the insulin signaling pathway.

I speculate the primary origin of these limitations is the small, high-school scale of this project. Gual et al.'s 2003 study is one of many administering insulin to 3T3-L1 cells that utilized a platform of 100 mm petri dishes, growing the fibroblasts in these dishes before the differentiation process and assay; far fewer studies utilize 6-well plates. These petri dishes have a surface area of 56.7 cm² compared to the 9.6 cm² platform of 6-well plates (Thermofisher, n.d.). More cells would be present in each sample, resulting in a larger amount of protein; more proteins would allow for all samples to be successfully quantified using the Western blot, helping gain a more representative idea of the data for all groups.

Additionally, during a Western blot, cell pellets must first be lysed to release all internal DNA and proteins, and proteins are then isolated by creating a protein pellet and aspirating away the DNA. However, due to the small size of the cell pellets, certain samples (including Insulin Control 1 and 2) had difficulty in the separation of proteins and DNA, as protein amounts were so small that DNA continued to adhere to the pellet. The presence of DNA in a Western blot can lead to poor quality band output, which could be the cause of the lack of data for several Akt samples. Further, I could only analyze 3 samples per group, which limited the representativeness of the data gathered; with more samples per group, a more accurate sense of the results could be found.



Another potential source of error throughout the experiment was the process of differentiation. While protocols from Zhao et al. (2019) were meticulously followed, the resources of my high school lab did not permit a means of confirming differentiation aside from visual appearance. As previously mentioned, fibroblasts appear more narrow and stretched while adipocytic cells are rounder and brighter under a brightfield microscope; wells of the 6-well plates were considered ready for experimentation after the majority of cells expressed this appearance. However, because I did not run an assay to fully confirm successful differentiation, it is possible that some samples contained more fully differentiated adipocytes than others; since different cell types contain different amounts of proteins, the presence of fibroblasts in wells thought to have differentiated into adipocytes could have interfered with some results.

Future Steps

3T3-L1 cells should be directly cultured as fibroblasts in a larger seeding platform such as 100 mm petri dishes prior to differentiation. After the differentiation process, an assay assessing presence of fat cells should be performed, such as a Nile Red or Oil Red O Stain; by staining and quantifying the amount of lipid droplets present in a sample of L1 cells, future studies could affirm whether differentiation methods were successful (Rizzatti et al., 2013). Once cells have differentiated, pretrials should utilize a wider variety of carbaryl concentrations (100–500 nM) to identify the effects of more tolerable concentrations compared to more drastic ones. With more protein collected, DNA and protein separation should not continue to be an issue; however, future research should perform several Western blots to analyze more samples of each group. Similar to the Insulin Control which was administered only insulin, another control group should be created that administers only carbaryl to cells to see how it impacts baseline cells; comparing this to cells that were exposed to carbaryl but later treated with insulin would indicate whether carbaryl interferes with protein levels in the insulin signaling pathway prior to or after its activation. Finally, to determine whether carbaryl affects the insulin signaling pathway before insulin reception or after, p-Akt should be analyzed, as its decrease after exposure to carbaryl could potentially indicate that carbaryl directly blocks reception of insulin.

Contribution to Existing Literature and Research Gap

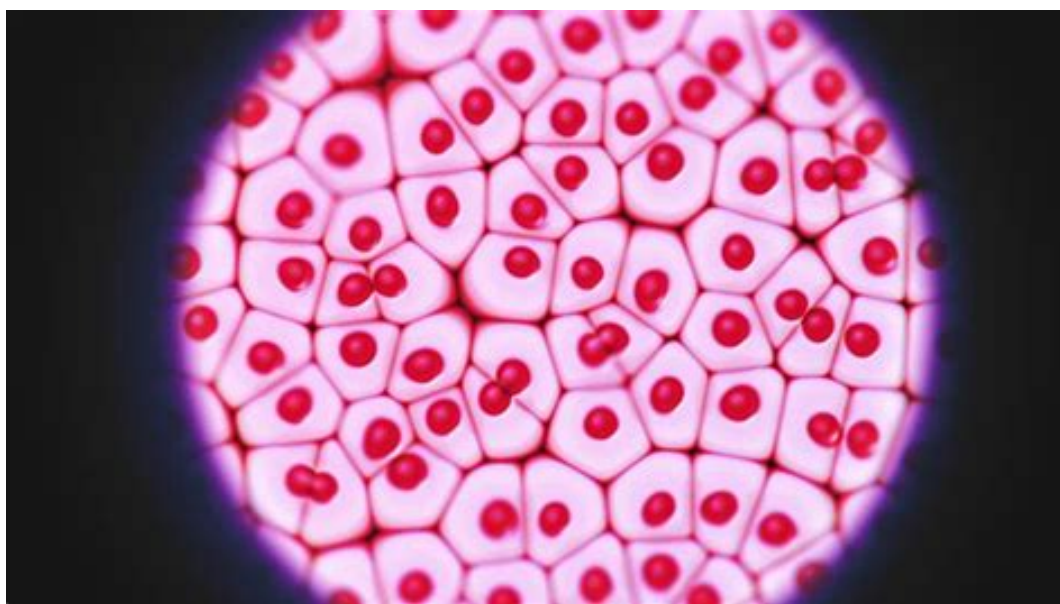
This study has contributed to the existing academic conversation by directly administering carbaryl to 3T3-L1 adipocytes in-vitro with the purpose of identifying the chemical's effects on the insulin signaling pathway of fat cells. It was found that carbaryl treatment correlates with lower detectable amounts of essential proteins in the insulin signaling pathway, thus negatively impacting the efficiency of the insulin signaling pathway after its stimulation. Sahana & Agarwal (2018) had suggested that carbaryl might impact the function and effectiveness of this pathway, a proposition that has gained evidence through this study. While the lack of data on carbaryl's effect on p-Akt limits possible conclusions on whether carbaryl docks directly at insulin receptors as implied by Popovska-Gorevski et al. (2014), these results support the idea that carbaryl has a measurable effect on insulin reception, bridging a previously existing gap in research about the effects of this pesticide chemical on insulin signaling in mammal fat cells.

Implications



The results of this study indicate that exposure to carbaryl leads to lower amounts of Akt present in cells, consequently decreasing the pathway's efficiency in lowering blood glucose levels because Akt is the enzyme responsible for signaling downstream proteins that open the Glu4 channels in fat cell membranes to lower the amounts of glucose in the blood and instead utilize it for energy and lipid storage. If the insulin signaling pathway's function is diminished, the body is more susceptible to sustained high levels of glucose in the blood, the primary marker of insulin regulation conditions such as Type 2 Diabetes.

As demonstrated by He et al. (2020), insulin resistance can commonly occur or become heightened by environmental factors, notably including pesticides. Yet carbaryl is one of the most widely used chemicals in pesticides – not only in agriculture, but in residential and commercial settings as well. This means it's often exposed to humans, pets, and other non-target mammalian organisms. Since this study provides evidence to show that carbaryl is a strong potential contributor towards the development of insulin resistance conditions, to patients already susceptible to or developing Type 2 Diabetes, an awareness of these environmental factors must be emphasized. Future research must build upon these findings to determine whether a need for caution is necessary surrounding the use of carbaryl in areas of high mammal contact such as houses and gardens.



PROVING KEPLER'S LAW

BY IBRAHIM CHISHTI





INTRODUCTION

The word “Planet” comes from ancient Greek “Planetai” meaning wanderer. This is because compared to the backdrop of stars they appear to wander independently. Over thousands of years, humanity has come to quantify this wandering and now has mathematical tools to calculate and predict their motions ahead of time. A vital step in finding these rules was the discovery (and proof) of Kepler’s Laws, a task so difficult it forced Newton to invent calculus.

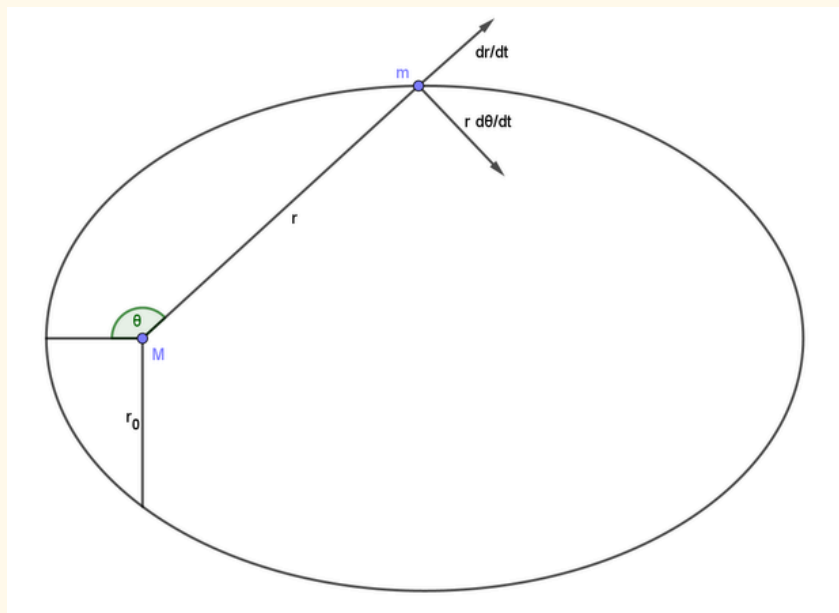
KEPLER’S 1ST LAW

Kepler’s 1st law details the shape of an orbit. It states that the shape of an orbit will always be an ellipse when only 2 bodies are involved. This assumption explained the slight deviations of the other planets from circular paths as well as the recurrence of Halley’s comet. This went unproven for some time until Newton presented a proof for the shape of an orbit.

Within each ellipse there are two special points called foci, the ellipse is defined as the locus of all points such that the sum of the distances to the two foci is constant. In the 1st law, they also provide another purpose, one focus being the position of the parent body while the other must remain empty.

It was later found that the 1st law follows from Newton’s law of gravitation, which showed that the force exerted by an object due to gravity is inversely proportional to the square of the distance.

Proof of the 1st Law



For the purposes of this proof, we will use the above diagram.

We can now write equations for both conservation of energy and conservation of angular momentum.



For conservation of energy, we can use the inverse square law to find a value for the GPE of one body moving around another. Combining this with the kinetic energy gives us.

$$E = \frac{1}{2}mv^2 + \int \frac{GMm}{r^2}dr \Leftrightarrow E = \frac{1}{2}mv^2 - \frac{GMm}{r}$$

Note that we can decompose v in terms of $dr/d\theta$ and $r d\theta/dt$, giving us:

$$E = \frac{1}{2}m \left(\left(\frac{dr}{d\theta} \right)^2 + \left(\frac{d\theta}{dt} \right)^2 \right) - \frac{GMm}{r}$$

And likewise for angular momentum gives us:

$$L = mr^2 \frac{d\theta}{dt}$$

We can now make the substitution of $\rho=1/r$ and rearrange to give:

$$\theta = \int \frac{L\rho}{m} \frac{dt}{d\rho} d\rho$$

Note that:

$$\frac{dr}{dt} = - \frac{1}{\rho^2} \frac{d\rho}{dt}$$

So:

$$\theta = - \int \frac{L}{m \frac{dr}{dt}} d\rho$$

From our conversation of Energy Equation, we know that:

$$\left(\frac{dr}{dt} \right)^2 = \frac{2E}{m} + 2GM\rho - \frac{L^2}{m^2} \rho^2$$

Using the substitutions:

$$r_0 = \frac{L^2}{GMm^2}$$

We can find that

$$\frac{dr}{dt} = \frac{L}{m} \sqrt{\frac{e^2}{r_0^2} - \left(\rho - \frac{1}{r_0} \right)^2}$$

Substituting to our equation for θ from before, gives us

$$\theta = - \int \frac{1}{\sqrt{\left(\frac{e}{r_0} \right)^2 - \left(\rho - \frac{1}{r_0} \right)^2}} d\rho$$

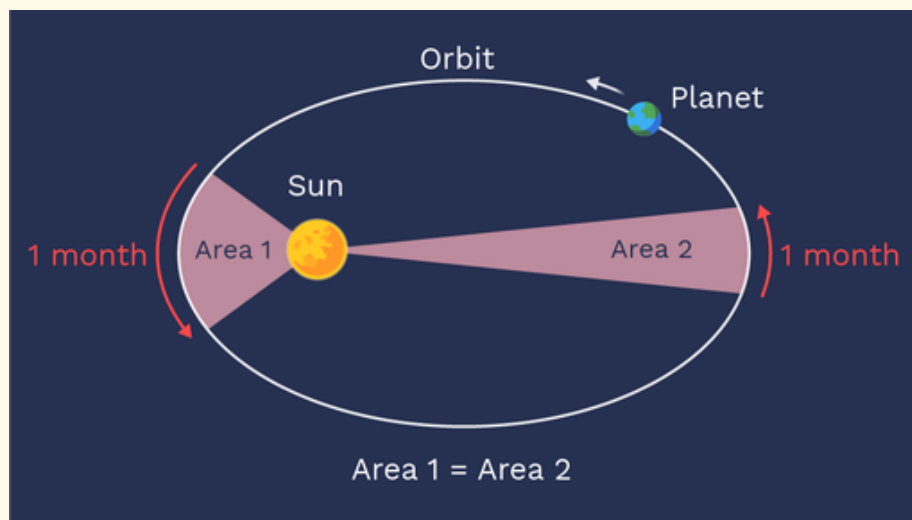


This is exactly the polar equation for an ellipse with the origin at one focus. This gives us the following law:

“Each planet's orbit about the Sun is an ellipse. The Sun's center is always located at one focus of the orbital ellipse.”

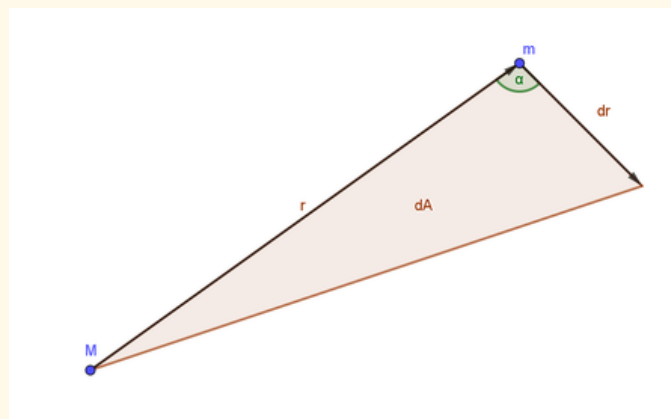
KEPLER'S 2nd LAW

As a planet gets further away from the Sun, it appears to slow down. For example, on approach to Earth, Halley's comet appears to go extremely quick, however on its way back to the outer solar system it slows down massively. From an energy perspective this makes sense, since the further away an object is from the sun, the more Gravitational Potential Energy it has, this means that the Kinetic Energy must decrease for the total energy to be conserved. However Kepler also noticed a key detail in the speed of a planet as it gets further from the Sun.



Looking at the area swept out by the imaginary line between the planet and its star, the area it sweeps out in a given time seems to be constant. It could also be said that the rate of change of area swept out is constant as well.

Proof of the 2nd Law





The diagram above shows the area swept out by a line conjoining one body to its orbital parent in some small time dt . We can use the Area Sin rule to find

$$dA = \frac{1}{2} r dr \sin(\alpha)$$

We can write this as a cross product, assuming A is a vector perpendicular to the plane of the triangle, giving us

$$dA = \frac{1}{2} r \times dr$$

Therefore the rate of change of Area is

$$\frac{dA}{dt} = \frac{1}{2} r \times \frac{dr}{dt}$$

To show Kepler's second law, we need to show that d is constant.

To do this, we can differentiate both sides with respect to time, giving us

$$\frac{d^2 A}{dt^2} = \frac{d}{dt} \left(\frac{1}{2} r \times \frac{dr}{dt} \right)$$

Using the product rule for differentiation gives us

$$\frac{d^2 A}{dt^2} = \frac{1}{2} \left(\frac{dr}{dt} \times \frac{dr}{dt} + r \times \frac{d^2 r}{dt^2} \right)$$

The first term on the right hand side is a vector with a cross product to itself, this must equal 0. The second term is $r \times d^2 r/dt^2$. Note that $d^2 r/dt^2$ is just the acceleration of the body. By Newton's law of gravitation this must be directed towards the body, in the same direction as r . This means that $r \times d^2 r/dt^2$ must also equal 0. Since all terms on the right hand side are equal to 0, the left hand side must also equal 0. Therefore

$$\frac{dA}{dt} = \text{constant}$$



From this we can conclude that:

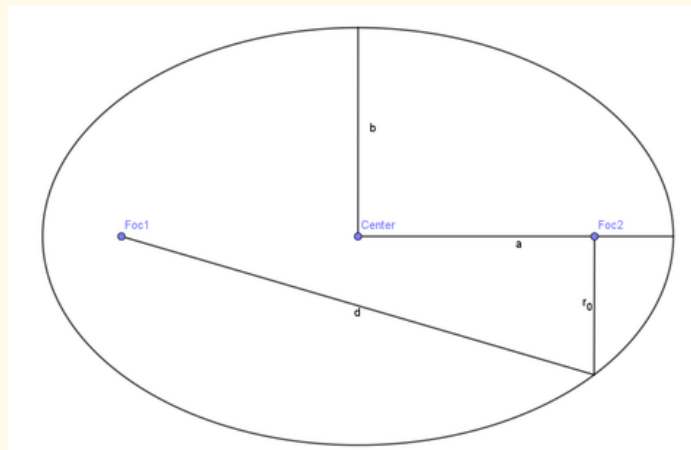
“The imaginary line joining a planet and the Sun sweeps equal areas of space during equal time intervals as the planet orbits.”

KEPLER'S 3rd LAW

The third law relates the size of the orbit to the time it takes to complete it. Since we have already shown that an orbit is elliptical, we will use the semi-major-axis of the orbit as our metric for size. The semi-major-axis of an ellipse is just the longest dimension of the ellipse.

Kepler noticed that the square of the time taken to complete an orbit is proportional to the cube of the semi-major-axis of the orbit.

Proof of the 3rd Law



Consider the area of the orbit:

$$A = \pi ab = \pi a^2 \sqrt{1 - e^2}$$

(Note that $b = a\sqrt{1 - e^2}$ where e is the eccentricity of the ellipse)

We know from Kepler's 2nd law that the rate at which Area is swept out is:

$$\frac{dA}{dt} = \frac{1}{2} \mathbf{r} \times \frac{d\mathbf{r}}{dt} = \frac{1}{2} \mathbf{r} \times \mathbf{v} = \frac{L}{2m}$$

Where L is the planet's angular momentum.



The orbital period is just the time to sweep out the entire area, resulting in the following:

$$T = \frac{\pi a^2 \sqrt{1-e^2}}{\frac{L}{2m}} = \frac{m}{L} 2\pi a^2 \sqrt{1-e^2} T^2 = \frac{m^2}{L^2} 4\pi^2 a^4 (1-e^2)$$

Note that from the diagram above, the line between the two foci (of length $2a$) forms a right-triangle with r_0 and d . This, along with the definition of an ellipse telling us that $d+r_0=2a$ gives us the following by Pythagoras:

$$(2a - r_0)^2 = 4a^2 e^2 + r_0^2$$

We also already defined r_0 in the proof of Kepler's first law, so using that we get:

$$\frac{m^2}{L^2} = \frac{1}{GMa(1-e^2)}$$

Substituting this into our equation from before gives us:

$$T^2 = \frac{m^2}{L^2} 4\pi^2 a^4 (1-e^2)$$

Therefore we can now say:

"The squares of the orbital periods of the planets are directly proportional to the cubes of the semi-major axes of their orbits."

CONCLUSION

When the distances to the planets were discovered and their motion carefully monitored. It was found that Uranus was behaving in unexpected ways. It was assumed that another planet must have been tugging on Uranus and altering its motion. This planet was then found and later named Neptune. A similar phenomenon appeared to happen with Mercury too. Its orbit would precess at a very small but measurable rate. It would appear we've had this problem before, the position of the new planet was calculated, the name "Vulcan" had been put forward and astronomers searched only to find nothing.

It was later discovered that this deviation from Kepler's laws was not due to an undiscovered planet, but ultimately due to a faulty theory of gravity. Einstein's theory of general relativity cleared up the discrepancy, but not before it disproved all of Kepler's laws. Despite this they can still be used in most circumstances and they ultimately served their role in giving humanity a better understanding of the cosmos.

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